Measurement of $\nu_\mu + \bar{\nu}_\mu$ Charged-Current Single Pion Production Differential Cross Sections on Argon with the MicroBooNE Detector in the NuMI Beam

Analysis note for the FHC, RHC and combined configurations of the full NuMI exposure

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Abstract

We describe the analysis chain for the measurement of flux-averaged $\nu_\mu + \bar{\nu}_\mu$ charged-current single-charged-pion ($\mathrm{CC}\pi^\pm$) cross sections on argon with the MicroBooNE detector in the NuMI beam. The signal region is blind, and every cross section quoted is extracted from central-value pseudo-data. The signal is one muon and one charged pion, any number of nucleons and no neutral pion, with $p_\mu > 0.15\,\mathrm{GeV}/c$, $p_\pi > 0.175\,\mathrm{GeV}/c$ and a muon–pion opening angle below 2.6 rad. The cross sections are averaged over the $\nu_\mu + \bar{\nu}_\mu$ flux integrated above 60 MeV, which includes the part of the spectrum below the pion-production threshold and so lowers the flux-averaged value.

The full NuMI exposure is analysed in three configurations: forward horn current (FHC, 7.766×10^{20} POT), reverse horn current (RHC, 11.082×10^{20} POT) and their combination (18.848×10^{20} POT). The selection requires a muon candidate and exactly one contained charged-pion candidate and reaches 18.3–18.4% efficiency at 57% purity.

The results include single differential cross sections $d\sigma/dp_\mu$, $d\sigma/d\cos\theta_\mu$, $d\sigma/d\cos\theta_\pi$, $d\sigma/d\theta_{\mu\pi}$ and the one-bin total cross section for the three beam configurations, and the double differential cross section $d^2\sigma/d\cos\theta_\pi\, d\cos\theta_\mu$ only for the combined configuration. The differential results are extracted by Wiener-SVD unfolding and provided with their additional-smearing matrices A_C and a covariance that contains the flux, interaction-model, re-interaction, detector, muon-momentum-scale, normalisation and statistical terms. The total cross section carries a 34–36% uncertainty, dominated by the flux (28–30%, the 17% flux normalisation amplified by the background subtraction) and the detector (14%).

The extraction passes closure, ensembles of pseudo-experiments (pull widths 0.92–1.04) and pseudo-data from reweighted interaction models (largest bias 1.2σ, in one sparse bin); no closure test with an independent generator has been made.

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1 Introduction

Charged-current single charged-pion (CC$\pi^{\pm}$) production,

$$\nu_\mu(\bar{\nu}_\mu) + \mathrm{Ar} \longrightarrow \mu^{\mp} + \pi^{\pm} + X, \tag{1}$$

is among the most frequent neutrino interactions at energies of 1–5 GeV and a significant background to oscillation measurements at the same energies, in particular to ν_e appearance at NOvA [3] and T2K [4] and at the future DUNE [5]. The cross section on nuclear targets is poorly constrained: the GENIE v3, NuWro and NEUT generators differ by 20–30% in absolute rate and show shape differences in the individual kinematic variables.

MicroBooNE is a liquid argon time-projection chamber (LArTPC) [28] at Fermilab that recorded data from 2015 to 2021, on axis in the Booster Neutrino Beam and off axis in the NuMI beam. The NuMI exposure contains neutrino- and antineutrino-enhanced samples with a softer and broader spectrum than the Booster beam. We measure CC$\pi^{\pm}$ cross sections on argon with the full NuMI exposure, using Pandora pattern recognition [6, 12] and Wiener-SVD unfolding [7].

1.1 Measurement overview

The two horn polarities are analysed as independent measurements, FHC and RHC, each with its own exposure and integrated flux, and as their combination; the signal, the phase space and the conventions are defined in Section 2.

Figure 1 shows the extraction. The same selection is applied to the beam data, to the beam-off data that measure the cosmic-ray background, and to the simulation. The simulated neutrino background, the beam-off data and the simulated interactions outside the cryostat are subtracted, and the remaining spectrum is unfolded with the simulated response. The result of each extraction is a cross section, the additional-smearing matrix A_C that describes the bias of the regularised unfolding, and the covariance matrix of the uncertainties.

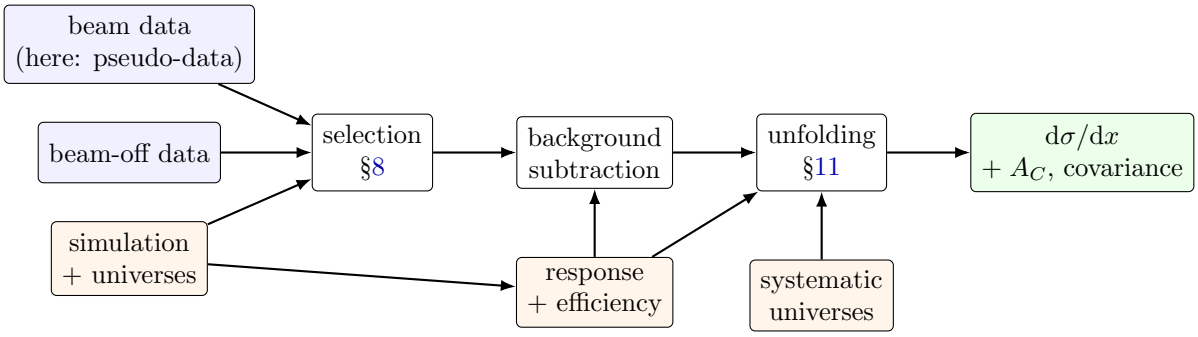


Figure 1: The extraction chain. The same selection is applied to data, beam-off data and simulation; the simulation also supplies the response and the systematic universes. The output is a cross section together with the additional-smearing matrix A_C and the covariance, all three of which are needed to compare the result with a prediction.

Five differential distributions and the total cross section are measured in each configuration: $d\sigma/dp_\mu$, $d\sigma/d\cos\theta_\mu$, $d\sigma/d\cos\theta_\pi$, $d\sigma/d\theta_{\mu\pi}$, and the cross section in three regions of pion momentum. The coarse three-region form of p_π follows from the saturation of the range-based pion momentum near 0.34 GeV/c (§14.3). The differential results estimate A_C times the true spectrum, so a prediction is compared after multiplication by A_C (§14.4); A_C does not preserve the normalisation, and the total cross section is taken from the one-bin extraction, for which $A_C = 1$ (§14.6).

The signal region is blind. All results are extracted from pseudo-data, Poisson fluctuations of the central-value simulation at the recorded exposure. The four background control regions

are mostly disjoint from the signal region; they have been studied with pseudo-data, and the procedure for their comparison with beam-on data is presented in §10.

For clarity, we document the analysis in three documents. This note defines the measurement and describes the selection, the extraction, the uncertainties, the validation and the inclusive results. The proton-tagged subsample, which adds hadronic-mass and transverse-kinematic-imbalance observables, is analysed in a companion note [45], which refers to this note for everything the two share. The technical supplement gives the final configuration in one place, the studies behind each choice and the alternatives that were rejected.

The separation of the documents represents a possible publication strategy: two PRD papers or a PRL and companion PRD.

2 Measurement definition

Table 1 defines the measurement. Every result of this note refers to this definition; the sections given in the last column describe each element.

Table 1: Definition of the inclusive measurement.

Element	Definition	Section
Interaction	ν_μ or $\bar{\nu}_\mu$ charged current on argon; $\nu_\mu \to \mu^-\pi^+$ and $\bar{\nu}_\mu \to \mu^+\pi^-$ are both signal (charge-blind)	6
Final state	Exactly one muon and exactly one charged pion among the primary particles; no neutral pion, kaon or other meson; any number of protons and neutrons	6
Coherent production	Part of the signal	6
Vertex	True vertex in the fiducial volume $10 < x < 246$, $-101 < y < 101$, $10 < z < 986$ cm, excluding $675.1 < z < 775.1$ cm	7
Phase space	$p_\mu > 0.15\,\mathrm{GeV}/c$, $p_\pi > 0.175\,\mathrm{GeV}/c$, $\theta_{\mu\pi} < 2.6$ rad	6
Neutrino direction	Angles and transverse quantities about the neutrino direction: the true direction of each event at truth level, the NuMI target-to-vertex direction at reconstruction level (28.6° from the detector z axis)	4.2
Flux	$\Phi_{\nu_\mu} + \Phi_{\bar{\nu}_\mu}$ of the new-G4 PPFX central value, integrated above 60 MeV: FHC 6.81159, RHC 6.44646, combined 6.59691×10^{-10} cm^{-2} POT^{-1}; the threshold includes the flux below the pion-production threshold	4.4
Exposure	FHC 7.766, RHC 11.082, combined 18.848×10^{20} POT; each run normalised to its own recorded exposure	5
Target	$N_{\mathrm{Ar}} = 8.710 \times 10^{29}$ argon nuclei in the fiducial volume	4.4
Normalisation	Cross section per argon nucleus, 10^{-38} cm^2/Ar; $d\sigma/dx\|_j = \hat{N}_j/(\Delta x_j\,\Phi\,N_{\mathrm{Ar}})$	11
Unfolding	Wiener-SVD; each differential result estimates $A_C\,x_{\mathrm{true}}$, and a prediction is compared after multiplication by A_C; the one-bin total has $A_C = 1$	14.4
Observables	p_μ (6 bins), p_π (3 regions), $\cos\theta_\mu$ (8), $\cos\theta_\pi$ (5), $\theta_{\mu\pi}$ (5), one-bin total; FHC, RHC and combined. $\cos\theta_\pi \times \cos\theta_\mu$ (2×3), combined only	App. A
Covariance	Flux, interaction model, re-interaction, detector, muon momentum scale, POT, target count, simulation, beam-off and data statistics	12

3 Physics context

3.1 Production mechanisms and nuclear effects

Between 0.5 and 5 GeV, charged-current single charged-pion production proceeds mainly through baryon resonances, dominated by the $\Delta(1232)$ and modelled with Rein–Sehgal-type form factors [30, 31, 32, 33, 34]. A non-resonant continuum and deep-inelastic scattering together contribute about 20% of the rate in the present phase space and are added incoherently by every generator used here. Coherent production off the whole nucleus contributes at most 10%; it

satisfies the topological signal definition of §6 and is part of the signal. In argon, Fermi motion, Pauli blocking and binding modify the kinematics, and final-state interactions (pion absorption at 25–30%, charge exchange and quasi-elastic scattering) move events between the CC$\pi^{\pm}$, CCπ^{0} and CC0π topologies at the 30% level. For this reason the signal is defined by the final-state particle content rather than by the production process. The same mixing makes CC0π the largest background: pion absorption, and a proton reconstructed as the pion, bring such events into the selection (Sec. 19 of the technical supplement).

3.2 Generators

Four generators are compared with the results; the model choices relevant to this measurement are listed in Table 2. The response and the pseudo-data are built from the MicroBooNE tune of GENIE v3 [16]. In the measured phase space the standalone generators span a factor of 1.6 in the total cross section, from GENIE (lowest) to NEUT (highest) (Table 19 of the technical supplement). Since the present results are extracted from pseudo-data, the model comparisons test the procedure; the spread indicates the size of the model differences the measurement is expected to resolve.

Table 2: Model content of the generators compared, restricted to what matters for interpreting this measurement.

Generator	Resonance model	Non-resonant / DIS	Final-state interactions
GENIE v3 [9, 20]	KLN–BS, 18 resonances to $W \lesssim 2$ GeV	Bodek–Yang, incoherent; AGKY transition	INTRANUKE hA (one effective interaction)
NuWro [21]	$\Delta(1232)$ only (Paschos–Yu)	separate amplitude; smooth transition to Bodek–Yang	Salcedo–Oset (in-medium Δ, two-body absorption)
NEUT [35, 23]	modified Rein–Sehgal, 18 resonances, Graczyk–Sobczyk Δ form factors	GRV98 with Bodek–Yang above $W \approx 2$ GeV	step-by-step intranuclear cascade
GiBUU [22]	Δ and higher resonances with in-medium spectral functions	PYTHIA fragmentation above $W \approx 1.6$ GeV	transport equations with pion optical potential

3.3 Previous measurements

Table 3 lists published CCπ^{+} measurements on nuclear targets. The two most relevant are from MicroBooNE. The Booster-beam measurement [44], on the same detector with an on-axis spectrum peaked at 0.7 GeV and a negligible $\bar{\nu}_{\mu}$ component, gives $\sigma = (3.75 \pm 0.07_{\text{stat}} \pm 0.80_{\text{syst}}) \times 10^{-38}$ cm^2/Ar and finds all tested generators above the data at very forward muon angles. The NuMI $\nu_e + \bar{\nu}_e$ CCπ^{+} measurement [43] uses the same beam, reconstruction and extraction framework and conventions as this analysis (Sec. 18.1 of the technical supplement). The NuMI flux at MicroBooNE is softer and broader than the Booster flux and has a 35% $\bar{\nu}_{\mu}$ component in FHC, and the flux-averaged cross section divides by the whole flux above 60 MeV while pion production starts near 0.3 GeV; the value of about 1.0×10^{-38} cm^2/Ar found here is therefore not directly comparable with the Booster result (§4.4). Among other experiments, the T2K off-axis measurement on carbon [41] has the most similar spectrum.

Table 3: Selected CCπ^+ cross-section measurements on nuclear targets. $\langle E_\nu \rangle$ is the approximate mean or peak neutrino energy. "Type" indicates whether a total integrated (T) or differential (D) cross section was reported. "Target" gives the nucleus primarily used for cross-section extraction.

Experiment	Target	$\langle E_\nu \rangle$ (GeV)	Type	Key result / reference
ANL 12-ft [36]	D	1.0	T	Bubble chamber; first σ vs E_ν
BNL 7-ft [37]	D	1.5	T	Higher statistics; 15% normalisation tension with ANL
K2K [38]	C	1.3	T	First accelerator CCπ^+ on carbon near detector
MiniBooNE [39]	CH_2	0.7	D	Q^2, W, p_μ, θ_μ; 20% excess over NUANCE
MINERvA [40]	CH	3.6	D	p_π, θ_π, Adler angles; NuMI medium-energy
T2K ND280 [41]	CH	0.85	D	p_μ, $\cos\theta_\mu$; off-axis beam similar to this work
ArgoNEUT [42]	Ar	9.6	T	First CCπ^+ on Ar; 115 ν + 158 $\bar{\nu}$ events (1.25×10^{20} POT)
MicroBooNE BNB [44]	Ar	0.7	D	Full BNB dataset (1.11×10^{21} POT); $\sigma = (3.75 \pm 0.80) \times 10^{-38}$ cm^2/Ar
MicroBooNE NuMI ν_e [43]	Ar	0.73	D	First $\nu_e + \bar{\nu}_e$ CCπ^+ on Ar; $\sigma = (0.93 \pm 0.28) \times 10^{-39}$ cm^2/nucleon

4 Beam, flux and exposure

4.1 NuMI beam and flux composition

The NuMI beam is produced by 120 GeV protons from the Fermilab Main Injector on a graphite target. Two magnetic horns focus charged secondary hadrons of one sign. In forward horn current (FHC) the horns focus positive pions and kaons and the beam is predominantly ν_μ; in reverse horn current (RHC) they focus negative mesons and the beam is predominantly $\bar{\nu}_\mu$. MicroBooNE is located 134.6 mrad (7.71°) off the NuMI beam axis (§4.2), which places the flux peak between 0.5 and 2.5 GeV and narrows the spectrum. At this off-axis position and energy the sign selection of the horns is incomplete, and both modes contain a large wrong-sign component, largest in RHC.

The flux composition at the detector is taken from the new-G4 central-value flux histograms, integrated above $E_\nu = 60$ MeV. The threshold removes an unphysical accumulation near $E_\nu = 0$, below the reaction threshold, which the flux group removes in the same way. The muon-flavour fractions of these histograms reproduce the integrated fluxes of §4.4 exactly.

Table 4: NuMI flux composition at the MicroBooNE site by neutrino type, for the FHC and RHC horn modes (new-G4 central-value flux, integrated over $E_\nu > 60$ MeV, the same threshold as the integrals of Table 6; above 0.1 GeV the RHC split would read 42.0/55.2%). FHC is ν_μ-dominant; RHC is $\bar{\nu}_\mu$-dominant with a large wrong-sign ν_μ fraction. The combined row weights each mode by its fluence, $\Phi\times$POT (FHC 5.29, RHC 7.14×10^{11} cm^{-2}; Tables 5 and 6), giving a near-balanced $\nu_\mu/\bar{\nu}_\mu$ mix. The signal is charge-blind ($|\nu_{\text{pdg}}| = 14$), so both ν_μ and $\bar{\nu}_\mu$ contribute in every mode.

Horn mode	ν_μ	$\bar{\nu}_\mu$	ν_e	$\bar{\nu}_e$
FHC	63.4%	34.0%	1.7%	0.9%
RHC	44.2%	53.3%	1.4%	1.2%
Combined	52.4%	45.1%	1.5%	1.1%

The Package to Predict the FluX (PPFX) [8] corrections are applied event by event through the central-value weight stored in the analysis ntuples. The absolute normalisation of §4.4 uses the same new-G4 histograms.

4.2 Beam direction in detector coordinates

Two different angles describe the geometry of MicroBooNE in the NuMI beam (Fig. 2). The off-axis angle is a property of the beamline: the detector lies 91.1 m from the NuMI axis at 672.7 m downstream of the target, a baseline of 678.8 m and an off-axis angle of 7.71° (134.6 mrad).[1] This angle does not describe the orientation of the TPC.

The angle that enters the measurement is the direction of the neutrinos relative to the detector z axis. Averaged over interactions in the active volume, the true neutrino direction in detector coordinates is (0.4624, 0.0489, 0.8853), 28.6° from z. The detector z axis is therefore neither the direction from which the neutrinos arrive nor the NuMI beamline axis, which is itself 23.2° from z.

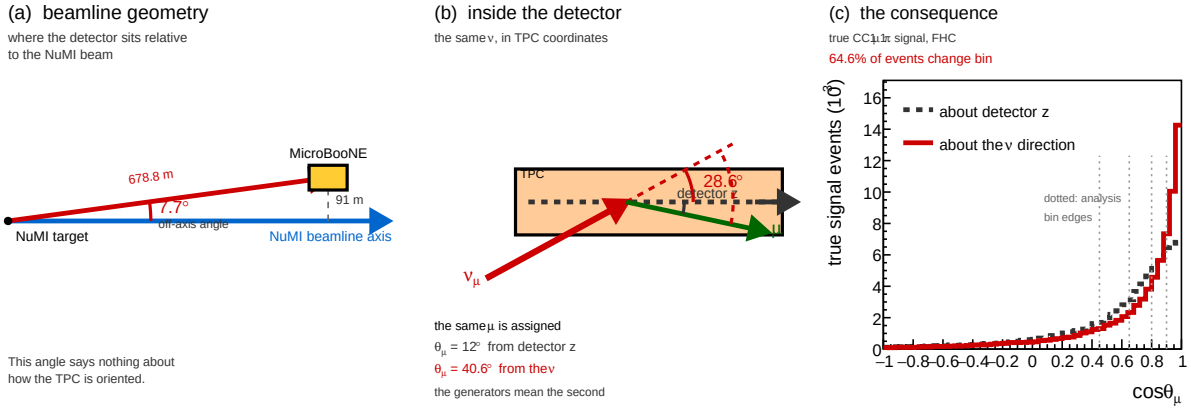


Figure 2: The NuMI geometry seen by MicroBooNE. (a) Beamline geometry, to scale: the detector sits 91.1 m off the NuMI axis at a baseline of 678.8 m, an off-axis angle of 7.7°. (b) The same neutrinos in TPC coordinates, to scale (the x–z projection; the quoted 28.6° is the full three-dimensional angle): the beam arrives 28.6° away from the detector z axis, so one and the same muon is assigned $\theta_\mu = 12°$ about z but $\theta_\mu = 40.6°$ about the neutrino. The generators use the second. (c) The resulting shift in the true $\cos\theta_\mu$ distribution; dotted lines mark the analysis bin edges. Under the two conventions 64.6% of signal events fall in a different bin (44.6% for $\cos\theta_\pi$).

[1] Computed from the detector position in beam coordinates, (5502, 7259, 67270) cm, with the rotation matrix used for the NuMI beamline geometry weights.

Every polar angle and every transverse-kinematic-imbalance variable is defined with respect to the neutrino direction. In the generator predictions the neutrino defines the $+z$ axis (the GENIE event records used here have $p_x^\nu = p_y^\nu = 0$ and $p_z^\nu = E_\nu$ exactly), so angles measured about the detector z axis would not correspond to the predictions they are compared with.

All angular and transverse observables are therefore measured about the neutrino direction, following the β convention of the NuMI ν_e CC1π analysis [1] and of the earlier NuMI ν_e measurement it builds on [2], which keeps the two NuMI measurements directly comparable. At truth level the exact neutrino direction of each event is used, which reproduces the generator convention. At reconstruction level the neutrino direction is approximated by the vector from the NuMI target, at (−31388, −3316, −60100) cm in detector coordinates (684 m from the detector centre and 678.8 m from the TPC origin), to the reconstructed interaction vertex.

The result is insensitive to this choice of reconstruction-level axis. A single fixed axis and the per-event β differ by a median of 2.4 mrad (at most 7.3 mrad), which moves 0.42% of muons across a $\cos\theta_\mu$ bin edge, compared with 64.6% for the detector-z convention. Against the true direction the two perform almost identically, with mean opening angles of 1.95° (β) and 2.00° (fixed axis). Neither recovers the divergence of the beam: the neutrinos come from decays along the whole decay pipe rather than from the target, and over a 684 m baseline a 2.5 m detector subtends about 4 mrad, so the vertex position carries little information on the direction. This residual is a resolution effect and is absorbed by the response matrix.

The muon and pion momenta, the muon–pion opening angle $\theta_{\mu\pi}$, the invariant mass $W_{\pi p}$ and all integrated cross sections do not depend on the choice of axis. Neither does the event selection, since no selection cut and no BDT input uses a polar angle.

4.3 Data exposure

The FHC measurement uses Runs 1, 2, 4 and 5 and the RHC measurement Runs 1, 2, 3, 4a, 4b and 4c; the combined measurement uses all fourteen overlay files. The exposure of each run period is listed in Table 5. For FHC Run 1 it is that of the data recorded without the open trigger, 2.192×10^{20} POT and 5 748 692 triggers (runs 4952–6748); the earlier open-trigger running (runs 3490–4950) is not used. Every run is normalised to its own exposure, so the prediction contains the run periods in the proportion of the recorded data.

The analysis is blind and the beam-on data of the signal region are not used. Each configuration is extracted from pseudo-data thrown run by run at that run's exposure, so every pseudo-data sample has the same run structure as the beam-on sample it stands in for. At unblinding the beam-on files of the same runs, with the same POT and trigger counts, replace the pseudo-data.

Table 5: Exposure by run period. "MC POT" is the simulated exposure of each period and "Data POT" the recorded exposure to which the pseudo-data of that period are thrown. Each run is normalised independently by Scale = (Data POT)/(MC POT), so the prediction is the data-weighted mixture of run periods, the normalisation a per-run beam-on dataset produces. The per-run factors range from 0.045 to 0.116; a single factor per horn mode would weight the runs by their simulated instead of their recorded exposure. The RHC Run-4 row sums the 4a, 4b and 4c files and the Run-3 row the aa–ae files; files of one run are scaled by the total simulated exposure of that run. Run 5 uses the ν_μ overlay produced with reweighted PPFX flux weights, the only Run-5 overlay available.

Run period	MC POT ($\times 10^{20}$)	Data POT ($\times 10^{20}$)	Scale
Run 1 (FHC)	23.282	2.192	0.0942
Run 2 (FHC)	24.934	1.268	0.0509
Run 4 (FHC)	28.335	2.075	0.0732
Run 5 (FHC)	19.300	2.231	0.1156
FHC total	**95.85**	**7.766**	*per run*
Run 1 (RHC)	8.997	0.605	0.0673
Run 2 (RHC)	57.865	2.591	0.0448
Run 3 (RHC, aa–ae)	55.185	5.003	0.0907
Run 4 (RHC, 4a–4c)	32.586	2.883	0.0885
RHC total	**154.63**	**11.082**	*per run*
Combined (FHC+RHC)	**250.49**	**18.848**	*per run*

With the full exposure, the data-statistical uncertainty is smaller than the flux and detector terms in every observable (Table 20 of the technical supplement–Table 22 of the technical supplement).

4.4 Integrated flux and target count

The absolute normalisation uses the new-G4 NuMI flux integrated above $E_\nu = 60$ MeV. The signal does not distinguish the charge of the lepton ($|\nu_{\mathrm{pdg}}| = 14$ covers both $\nu_\mu \to \mu^- \pi^+$ and $\bar{\nu}_\mu \to \mu^+ \pi^-$), so the normalisation is the summed flux $\Phi = \Phi_{\nu_\mu} + \Phi_{\bar{\nu}_\mu}$. The RHC flux is $\bar{\nu}_\mu$-dominated and differs in size from the FHC flux, so each configuration is normalised with its own integral (Table 6).

Table 6: Integrated ν_μ and $\bar{\nu}_\mu$ fluxes per POT used by the analysis (new-G4 flux, $E_\nu > 60$ MeV), and the combined $\nu_\mu + \bar{\nu}_\mu$ value used as the per-configuration normalisation. The combined row is the data-POT-weighted average of the two modes (weights 7.766 : 11.082; Table 5). Each configuration is normalised with its own flux: the RHC flux is 5.4% below the FHC one, so using a single value for both would bias the RHC cross section by that amount.

Config	Φ_{ν_μ}	$\Phi_{\bar{\nu}_\mu}$	$\Phi_{\nu_\mu+\bar{\nu}_\mu}$	$\bar{\nu}_\mu$ frac.
	($10^{-10}\,\mathrm{cm}^{-2}\,\mathrm{POT}^{-1}$)			
FHC	4.43515	2.37644	6.81159	34.9%
RHC	2.92348	3.52298	6.44646	54.6%
Combined	3.54634	3.05057	6.59691	46.2%

The mean energy of the $\nu_\mu + \bar{\nu}_\mu$ flux above 60 MeV is 0.479 GeV (FHC) and 0.475 GeV (RHC). For ν_μ alone it is 0.468 and 0.544 GeV, for $\bar{\nu}_\mu$ alone 0.500 and 0.417 GeV; in RHC the antineutrinos are the focused component and are softer at this off-axis angle. Only 40% of the

flux lies above 0.3 GeV, 26% above 0.5 GeV and 13% above 1 GeV. Since the flux-averaged cross section divides by the whole flux above 60 MeV while single-pion production starts near 0.3 GeV, the total is about a factor 2.5 smaller than it would be for the Booster spectrum peaked at 0.7 GeV. This, together with the measured phase space ($p_\pi > 0.175$ GeV/c, $\theta_{\mu\pi} < 2.6$ rad), is why the value of about 1.0×10^{-38} cm^2/Ar found here differs from the 3.75×10^{-38} cm^2/Ar of the Booster measurement (§3.3).

Every simulated event carries the PPFX central-value flux weight (§5; mean 0.944 on selected signal), so the integrated flux that divides the event rate must be the PPFX-corrected central value. The central-value histograms of the new-G4 flux file are taken to be that central value. An integral of the uncorrected GEANT4 flux would lower every absolute cross section by 5–6%. The integrated values were validated in four independent ways: standalone NuWro and GENIE event generation on the same flux, the numbers of the flux author, and a GENIE closure.

The number of argon nuclei in the fiducial volume is

$$N_{\mathrm{Ar}} = \frac{\rho_{\mathrm{LAr}} \cdot V_{\mathrm{FV}} \cdot N_A}{A_{\mathrm{Ar}}} = 8.710 \times 10^{29}, \tag{2}$$

with $\rho_{\mathrm{LAr}} = 1.3836$ g/cm^3, $A_{\mathrm{Ar}} = 39.948$ g/mol and $N_A = 6.02214076 \times 10^{23}$ mol^{-1}. The fiducial box, $10 < x < 246$, $-101 < y < 101$, $10 < z < 986$ cm (4.6528×10^7 cm^3), minus the unresponsive region $675.1 < z < 775.1$ cm gives $V_{\mathrm{FV}} = 4.1761 \times 10^7$ cm^3; without the subtraction the target count would be 9.705×10^{29}. The extraction computes N_{Ar} from the same box and subtraction, and the value enters the conversion factor of §11 and the total of §14.6. The normalisation of each configuration is $N_{\mathrm{norm}} = \Phi_{\mathrm{mode}} \cdot N_{\mathrm{POT,mode}} \cdot N_{\mathrm{Ar}}$, with the flux and exposure of that mode (Tables 5 and 6).

5 Simulated and data samples

Neutrino interactions are simulated with the MicroBooNE tune [16] of GENIE v3 [9, 20] on argon-40, propagated through the detector with Geant4 [24], and overlaid on cosmic-ray data recorded without beam, so that simulated events contain the detector noise and cosmic activity of real events. Table 7 lists the samples.

Table 7: Input samples (final, full FHC+RHC). "n" is the number of overlay files; "MC POT" is the summed native simulated exposure ($\times 10^{20}$). The blind Poisson-thrown pseudo-data stands in for the (unused) beam-on data; the detector variations use the native Run-4 FHC and RHC samples. The per-file lists and POT and trigger values are in the configuration files of the analysis.

Sample	Composition	n	MC POT
ν_μ overlay (FHC)	Runs 1, 2, 4 (pandora) + Run 5 (reweighted-PPFX)	4	95.85
ν_μ overlay (RHC)	Runs 1, 2, 3, 4a, 4b, 4c (pandora)	10	154.63
Dirt MC (interactions outside the cryostat)	GENIE NuMI dirt overlays per run and horn mode (FHC Run 1 stands in for FHC Run 2, RHC Run 3b for RHC Runs 1 and 2)	7	n/a
EXT (beam-off)	cosmic data, per run, horn-mode-independent	7	n/a
Detector variations	Run-4 FHC + Run-4 RHC, CV + 8 knobs each	18	n/a
Pseudo-data (blind)	Poisson-thrown central value, per configuration	n/a	n/a
Beam-on data	blinded, not used until unblinding	n/a	n/a

Event weights. Each simulated event is weighted by

$$w = \mathrm{Scale[run]} \times w_{\mathrm{PPFX}} \times w_{\mathrm{tune}}, \tag{3}$$

where Scale[run] = (run Data POT)/(run MC POT) is the per-run factor of Table 5, w_{PPFX} the PPFX central-value flux correction and w_{tune} the GENIE tune weight. In the framework each overlay file receives the exposure of its run from the matching beam-on (here pseudo-data) entry, and the simulated exposure of a run is the sum of the distinct exposure values stored in its files, which is the run total under both file conventions in use. One rule is applied to every event weight,

$$w_{\mathrm{safe}} = \begin{cases} 1, & w \text{ non-finite, } w < 0, \text{ or } w > 30, \\ w, & \text{otherwise (a weight of exactly zero is kept),} \end{cases} \tag{4}$$

to the product $w = \mathrm{Scale} \times w_{\mathrm{PPFX}} \times w_{\mathrm{tune}}$, including the 0.65 normalisation weight of the dirt overlay that scales its generated exposure to the rate of interactions outside the cryostat expected at the NuMI site. The universes, the covariance, the unfolding inputs, the cut-flow histograms, the count-level tables, the control-region yields and transfer factors, and the beam-on comparison at unblinding all use Eq. (4), so that count-level and covariance-level numbers agree at the per-cent level.

Beam-off normalisation. Beam-off data contain cosmic rays only, so a beam-off sample is matched to a beam-on run by run period, independently of the horn polarity in which it was recorded. Each run period uses the beam-off samples of that period, scaled by the ratio of beam-on to beam-off triggers times the NuMI beam-occupancy factor 0.98. Several beam-off files of one run are pooled:

$$\mathrm{EXT}_{\mathrm{run}} = 0.98\, \frac{\mathrm{bnb_triggers[run]}}{\sum_{\mathrm{files}\in\mathrm{run}} \mathrm{ext_triggers}} \sum_{\mathrm{files}\in\mathrm{run}} N^{\mathrm{sel}}_{\mathrm{EXT}}. \tag{5}$$

No beam-off sample exists for Run 2; the Run-1 sample, which like Run 2 predates the cosmic-ray tagger, is used in its place and scaled by its own trigger count. The same beam-off files serve the FHC and RHC runs of a period, and the combined configuration lists each (file, run) pair once. The selected beam-off rate per 10^6 triggers is 1.75 (Run 1), 0.43 (Run 3b), 1.23 (Run 4) and 0.57 (Run 5), from 8, 14, 43 and 11 selected beam-off events, so the rates carry Poisson uncertainties of 15–35%. The Run-1 rate is 1.4 to 4 times the rates of the later periods, which is why the samples are matched by period instead of pooled. At the final cut the beam-off background is 24 events in FHC, 24 in RHC and 48 in the combination, 1.6%, 1.3% and 1.4% of the selected sample. Figure 3 shows the rates, scale factors and yields per period; every (beam-off file, run) pair enters each configuration once and that the trigger counts are consistent.

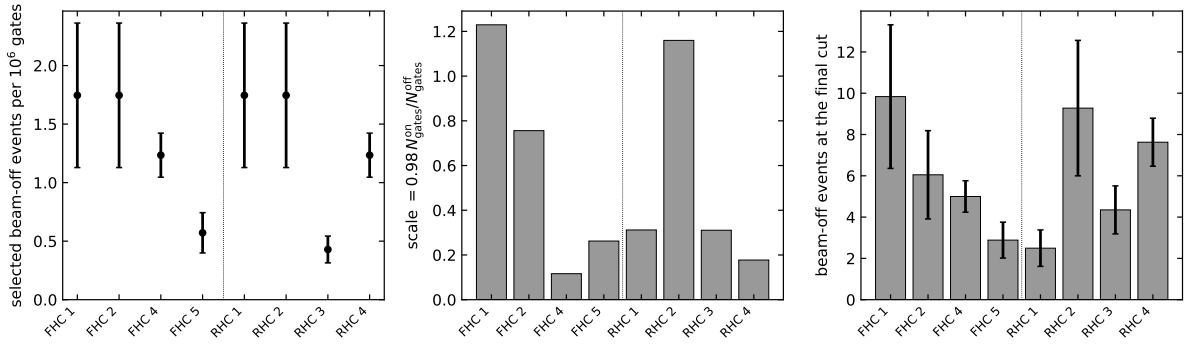


Figure 3: Beam-off normalisation by run period: selected beam-off rate per 10^6 triggers at the final inclusive cut, with its Poisson uncertainty (left), the scale factor 0.98×(beam-on triggers)/(beam-off triggers) (centre), and the resulting beam-off events at the final cut (right), for each beam-on run period of the FHC and RHC configurations. Run 2 of both modes uses the Run-1 sample. The Run-1 rate is 1.4 to 4 times the rates of the later periods.

Cosmic-ray rejection. No cosmic-ray rejection beyond the selection of §8 is applied. The beam-off contribution, 1.6% of the selected FHC prediction, is removed by a data-driven subtraction, so an additional cut could only reduce the statistical uncertainty of that subtraction, about five events in a sample of 1 522, and every variant studied removes more signal than that. The directional and calorimetric variants are described in Sec. 28 of the technical supplement.

Software trigger. Beam-on and beam-off files are filtered on the software trigger. The beam-off files of Runs 4 and 5 carry a trigger flag that is zero throughout, and for them the `swtrig_pre` flag is used instead. Simulated events that fail the software trigger are removed at the NeutrinoSlice cut; every simulated event that passes the topological cut also passes the trigger, so the requirement has no effect on the selected sample.

6 Signal definition

The signal is defined at generator level and is used to build the efficiency and the response; it is never applied as a reconstruction cut. It does not distinguish the charge of the lepton: $\nu_\mu \to \mu^- \pi^+$ and $\bar{\nu}_\mu \to \mu^+ \pi^-$ are both signal.

Neutrino flavour
$|\nu \text{ PDG}| = 14$ (ν_μ or $\bar{\nu}_\mu$).

Interaction type
Charged current.

True vertex
Inside the fiducial volume (§7).

Primary multiplicity
Among the primary final-state particles, exactly one muon ($|\text{PDG}| = 13$), exactly one charged pion ($|\text{PDG}| = 211$) of any momentum, no neutral pion ($|\text{PDG}| = 111$), no kaon ($K^\pm$, K^0, K_S, K_L) and no other meson. A second charged pion below the momentum threshold makes the event background. Pions from the decay of primary hyperons are not primary particles and are not counted. Any number of protons and neutrons is allowed.

Kinematic thresholds
$p_\mu > 0.15\,\text{GeV}/c$, $p_\pi > 0.175\,\text{GeV}/c$ and $\theta_{\mu\pi} < 2.6$ rad.

The thresholds define the measured phase space. The momentum efficiency rises sharply between 0.15 and 0.20 GeV/c, set by the 10 cm muon and 20 cm pion track-length requirements; below it the efficiency is 1–3% and the correction strongly model dependent. The upper bound on the opening angle removes the region above 2.6 rad, which is dominated by cosmic-ray and dirt background (about 7% purity in the $\theta_{\mu\pi} > 2.513$ rad bin). No lower bound is applied: the small-angle region has a purity of about 58%, so a cut there would remove signal without improving the purity. The same opening-angle requirement is applied at reconstruction level so that the signal and selection acceptances match. The first p_μ and p_π bins start, and the last $\theta_{\mu\pi}$ bin ends, at these thresholds.

Pion threshold. The NuMI ν_e CC1π analysis defines its pion signal by kinetic energy, $\text{KE}_\pi > 40$ MeV, equivalent to $p_\pi > 0.113\,\text{GeV}/c$. The threshold used here, $p_\pi > 0.175\,\text{GeV}/c$ ($\text{KE}_\pi > 84$ MeV), is higher because the pion momentum, reconstructed from the track range under the muon hypothesis and converted to the pion mass, degrades rapidly below it (Table 8). The pion-momentum bias is the signed quantity $b(p_\pi) = \langle (p_\pi^{\text{reco}} - p_\pi^{\text{true}})/p_\pi^{\text{true}} \rangle$, which has two regimes. Below the threshold the few pions that are reconstructed at all are biased upwards (+30 to

+131%): they are the tracks long enough to pass the selection. At the threshold the mass conversion removes most of the hypothesis bias (−8%, with 31% resolution). Above about 0.4 GeV/c the bias becomes increasingly negative, because pions interact before they stop and the visible range is truncated; the mean reconstructed momentum saturates near 0.34 GeV/c, and the bias reaches −65% above 0.8 GeV/c. Between 0.113 and 0.175 GeV/c the resolution is 51–56% and the selection efficiency 2.2%. A lower threshold would therefore add a bin with large, strongly model-dependent unfolding corrections, even as a single coarse bin, and it is not adopted; no cross section is reported below $p_\pi = 0.175$ GeV/c.

Table 8: Pion momentum reconstruction versus true p_π for correctly reconstructed signal pions. Below the 0.175 GeV/c threshold the bias and resolution grow and the reconstructed fraction halves.

True p_π [GeV/c]	Bias $\langle (p^{\text{reco}} - p^{\text{true}})/p^{\text{true}} \rangle$	RMS resolution	Reco fraction
0.080–0.113	+131%	73%	3.4%
0.113–0.150	+66%	56%	3.4%
0.150–0.175	+30%	51%	4.3%
0.175–0.250 (threshold)	**−8%**	**31%**	**10.0%**
0.250–0.400	−30%	21%	10.4%

7 Fiducial volume

The fiducial volume (FV) is

$$x \in [10.0,\ 246.0]\,\text{cm}, \quad y \in [-101.0,\ +101.0]\,\text{cm}, \quad z \in [10.0,\ 986.0]\,\text{cm}, \tag{6}$$

with the region of unresponsive wires, $z \in [675.1,\ 775.1]$ cm, excluded. Its volume is

$$\begin{aligned} V_{\text{FV}} &= (246 - 10) \times 202 \times (976 - 100)\,\text{cm}^3 \\ &= 236 \times 202 \times 876\,\text{cm}^3 \approx 4.17 \times 10^7\,\text{cm}^3. \end{aligned} \tag{7}$$

A larger containment volume, inset by 2 cm from the faces of the active volume and by 10 cm from the top face, is used to require that the pion-candidate track ends inside the detector:

$$x \in [2.0,\ 254.35]\,\text{cm}, \quad y \in [-113.53,\ +107.47]\,\text{cm}, \quad z \in [2.1,\ 1034.9]\,\text{cm}. \tag{8}$$

7.1 Use of the volumes

volume	extent	used for
fiducial (FV)	$[10, 246] \times [-101, 101] \times [10, 986]$ cm minus $z \in [675.1, 775.1]$ cm	vertex (true and reco)
containment	$[2, 254.35] \times [-113.53, 107.47] \times [2.1, 1034.9]$ cm	track end points
active TPC	$[0, 256.35] \times [-115.5, 117.5] \times [0, 1036.9]$ cm	reference only

The true and reconstructed fiducial volumes are the same box. The signal definition tests the true neutrino vertex and the selection the reconstructed vertex, through the same function, so both apply the exclusion of the unresponsive region. Vertex resolution and space-charge distortions can move a reconstructed vertex across a boundary; such events are part of the simulated response, true-in and reconstructed-out as inefficiency and true-out and reconstructed-in as the out-of-fiducial-volume background of Table 11.

The exclusion of the unresponsive region applies to the vertex only. An event with its vertex outside the region is kept even if one of its tracks crosses it; such tracks have degraded calorimetry and range, and the resulting inefficiency is part of the response. The region is also removed from the target count (Eq. (2)).

The fiducial volume is inset from the active TPC by about 10 cm in x, 15 cm in y, 10 cm at the upstream z face and 51 cm at the downstream face. The containment volume is inset by only 2 cm (10 cm at the top), since the range-based momentum and the pion-containment cut need the track to end inside the detector, not inside the fiducial region.

8 Event selection

The selection is implemented as a selection class of the extraction framework, which produces the response matrices and spectra of every result; an independent standalone implementation of the same cut sequence is kept as a cross-check. It consists of ten sequential cuts, with Cut 0 the unselected reference sample.

8.1 BDT-based particle identification

Three gradient-boosted decision trees (TMVA [26]), trained on the MicroBooNE overlay simulation, provide the calorimetric particle identification of the muon and pion candidates. All three use the same per-track inputs: the Bragg-peak dE/dx likelihoods under the proton, muon and minimum-ionising hypotheses, the log-likelihood-ratio (LLR) PID score, the Pandora track-shower score, and the space-charge-corrected track end points.

- **MIP BDT**: separates minimum-ionising tracks (muons, pions) from stopping protons. Both candidates must have a MIP score above -0.1 (Cuts 4 and 5).
- **Pion BDT**: a pion-versus-proton discriminant for the pion candidate, with a required score above -0.1 (Cut 5).
- **Muon BDT**: the same inputs plus the track length. It applies no threshold. The tracks that qualify as muon candidates are defined by the cuts of Table 9, and the qualifying track with the highest muon-BDT score is taken as the muon.

The two candidate-PID BDTs do not use the track length, so that their scores do not bias the momentum spectra.

8.2 Cut definitions

Cut 0. EventsInTrueFV. All events; the reference for truth-level studies.

Cut 1. NeutrinoSlice. At least one reconstructed track and the software trigger.

Cut 2. InFiducialVol. The space-charge-corrected reconstructed neutrino vertex lies inside the FV.

Cut 3. Topological. Topological score > 0.67. This Pandora multivariate score separates cosmic-ray from neutrino topologies and provides the main cosmic rejection: the beam-off yield falls by a factor of 28–29 between Cut 2 and Cut 3 (Table 11).

Cut 4. MuonCandidate. At least one primary track satisfies the muon criteria of Table 9. Of the qualifying tracks, the one with the highest muon-BDT score (§8.1) is the muon candidate; the track length enters only through the 10 cm requirement and as a BDT input.

Cut 5. ContainedPion. Exactly one primary track other than the muon candidate satisfies the pion criteria of Table 10 and ends inside the containment volume. The track with the highest LLR PID score is the pion candidate.

Cut 6. MuonIn3Planes. The muon candidate has at least one hit on each of the U, V and Y planes.

Cut 7. PionIn3Planes. The pion candidate has at least one hit on each of the U, V and Y planes.

Cut 8. ShowerCut. No primary electromagnetic shower, unless there is exactly one shower with fewer than 50 Y-plane hits more than 10 cm from the vertex, as expected for a delta ray or a low-energy nuclear photon.

Cut 9. OpeningAngle. The reconstructed muon–pion opening angle satisfies $\theta_{\mu\pi} < 2.6$ rad ($\approx 149°$), which removes the large-angle cosmic-ray and dirt background. The same requirement is part of the signal definition (§6).

Cut 10.
Nonprotons (final cut). Fewer than four primary tracks with LLR PID > 0.1 (non-proton-like), and fewer than five primary tracks in total.

Table 9: Muon candidate PID thresholds (applied at Cut 4).

Variable	Threshold	Description
Track score	> 0.80	Track-like PFP
Vertex distance	≤ 4.0 cm	Attached to ν vertex
Track length	> 10.0 cm	Minimum track length
LLR PID	> 0.20	Muon-like
MIP BDT	≥ -0.10	Minimum-ionising-particle

Table 10: Pion candidate PID thresholds (applied at Cut 5). Endpoint containment requires the track endpoint inside the containment volume. No requirement is made on the Bragg-pion score (Sec. 9.2).

Variable	Threshold	Description
Track score	≥ 0.50	Track-like PFP
LLR PID	> 0.10	Pion / non-proton-like
Vertex distance	≤ 4.0 cm	Attached to ν vertex
Track length	> 20.0 cm	Minimum track length
MIP BDT	> -0.10	MIP-like
Pion BDT	> -0.10	Pion-like (vs. proton)
Endpoint contained	yes	Inside containment volume

8.3 Kinematic reconstruction

Muon momentum. The muon momentum uses the range when the muon-candidate track ends inside the detector and multiple Coulomb scattering (MCS) [14] when it exits, since the range of an exiting track measures only the distance to the detector boundary. In the selected FHC sample 37% of the muons are contained and 63% exit, so the muon momentum is mostly measured by MCS: NuMI muons are energetic and the TPC is small. This impacts the resolution and the uncertainty, since the two estimators have different resolutions and systematic effects. The MCS estimator has a -10% bias and 30% resolution (Fig. 21), which enter through the response matrix. The detector variations do not change the MCS momentum scale directly; a dedicated momentum-scale term is evaluated in §12.3.

Pion momentum. The pion momentum starts from the range momentum under the muon hypothesis. Before a reconstructed p_π bin is assigned, it is converted to the pion hypothesis at fixed kinetic energy, $T = \sqrt{p^2_{\mu\text{-hyp}} + m^2_\mu} - m_\mu$ and $p_\pi = \sqrt{(T + m_\pi)^2 - m^2_\pi}$. The stored branch is unchanged; the conversion is part of the reconstructed-bin definition in the bin configuration, and it raises the reconstructed-to-true ratio of the lowest bin from 0.888 to 0.976 (Sec. 6.2 of the technical supplement). Because the muon mass (105.7 MeV) is close to the pion mass (139.6 MeV), the range-to-momentum conversion is a good estimate for a pion that stops, and it yields a momentum that is binned against the true momentum on the same axis. For a pion that interacts before stopping, the visible range is truncated and the estimate is biased low; this truncation, more than the mass hypothesis, limits the observable (§6). The kinetic energy from the proton-hypothesis range is not used: binned with the momentum edges it would shift the migration matrix systematically (a true 0.2 GeV/c pion has $KE \approx 0.104$ GeV, two bins lower, and a true 0.175 GeV/c pion would fall below the first reconstructed bin). The range momentum is only meaningful for contained tracks, which the containment requirement ensures.

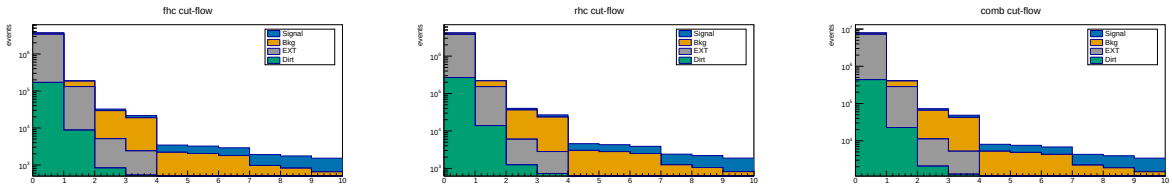
Angles. Polar angles are measured with respect to the neutrino direction $\hat{u}_\nu$: $\cos\theta_\mu = \hat{u}_\mu \cdot \hat{u}_\nu$ and $\cos\theta_\pi = \hat{u}_\pi \cdot \hat{u}_\nu$. For reconstructed quantities $\hat{u}_\nu$ is the β direction of Refs. [1, 2], from the NuMI target to the reconstructed vertex; at truth level it is the neutrino direction of the event (§4.2). The opening angle $\theta_{\mu\pi} = \arccos(\hat{u}_\mu \cdot \hat{u}_\pi)$ is computed from the reconstructed track directions and does not depend on $\hat{u}_\nu$.

8.4 Proton-tagged subsample

Requiring in addition a contained proton candidate with $p_p > 0.3\,\text{GeV}/c$ selects a proton-tagged subsample with one further cut. It gives access to hadronic-mass and transverse-kinematic-imbalance observables, with its own proton-assignment requirements and model dependence, and is described in the companion note (Sec. 2 of the proton-tagged note).

9 Selection performance and backgrounds

Figure 4 shows the cut-flow yields at full exposure for the three configurations, and Table 11 the yields by category after each cut. The final selection keeps 18.3% of the signal at 57.3% purity in FHC (18.4% and 56.6% in RHC, 18.4% and 56.9% combined).



(a) FHC. (b) RHC. (c) Combined.

Figure 4: Cut-flow yields at *full exposure* for FHC, RHC and combined (log scale; MC-prediction stack: signal blue, background orange, EXT grey, dirt green), with signal and background scaled per run to the recorded exposure. The detailed breakdown by category is in Table 11.

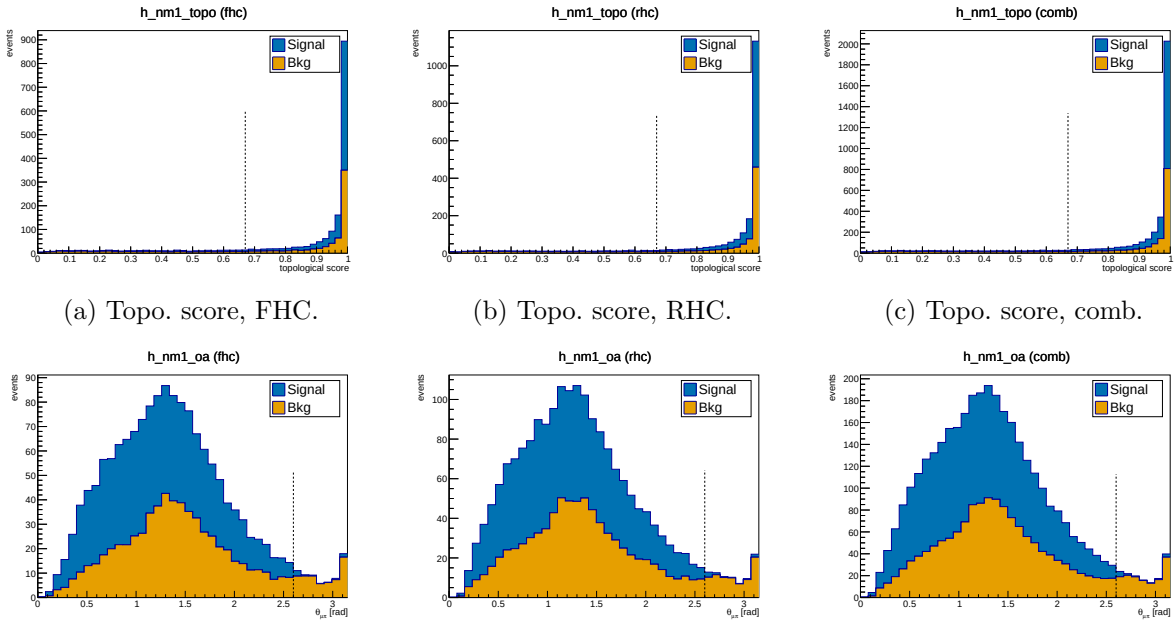
Table 11: Cut-flow yields at full exposure for FHC, RHC and combined, from the simulation (the expectation of the pseudo-data). The yields carry the full central-value weight, tune × PPFX × normalisation (§5, with Eq. (4)), as in the pseudo-data throw and the unfolding, so the final-cut yields agree with Table 27 to 1%. Signal and "Bkg" (all other neutrino interactions: out of FV, NC, CC1π^0, CC0π, CC$N\pi$, CCK and other CC) come from the ν_μ overlays scaled per run to the recorded exposure (Table 5); EXT is the beam-off data scaled by the per-run ratio of beam-on to beam-off triggers times 0.98, each run period using its own beam-off sample (§5); dirt is the GENIE NuMI dirt overlay. Pred. = Signal + Bkg + EXT + Dirt. Eff. is the cumulative signal efficiency relative to all generated signal (the None stage); Pur. = Signal/Pred.

Cut	Signal	Bkg MC	EXT	Dirt	Pred.	Eff. [%]	Pur. [%]
FHC (Runs 1,2,4,5; 7.766×10^{20} *POT)*							
None	4 753	235 281	3 256 689	170 169	3 666 891	100.0	0.1
InFiducialVol	3 955	52 835	122 860	8 738	188 387	83.2	2.1
Topological	2 995	24 262	4 299	827	32 383	63.0	9.2
MuonCandidate	2 689	16 484	1 870	530	21 573	56.6	12.5
ContainedPion	1 247	1 926	179	91	3 444	26.2	36.2
MuonIn3Planes	1 193	1 822	149	67	3 232	25.1	36.9
PionIn3Planes	1 110	1 642	116	38	2 906	23.3	38.2
ShowerCut	937	850	86	32	1 905	19.7	49.2
OpeningAngle	931	775	33	10	1 749	19.6	53.2
Nonprotons	**872**	**620**	**24**	**6.6**	**1 522**	**18.3**	**57.3**
RHC (Runs 1,2,3,4a,4b,4c; 1.108×10^{21} *POT)*							
None	5 778	286 540	3 560 975	267 449	4 120 741	100.0	0.1
InFiducialVol	4 828	65 474	139 122	13 907	223 331	83.6	2.2
Topological	3 743	30 446	4 767	1 243	40 199	64.8	9.3
MuonCandidate	3 363	20 709	2 090	730	26 892	58.2	12.5
ContainedPion	1 554	2 695	195	128	4 572	26.9	34.0
MuonIn3Planes	1 491	2 543	162	100	4 295	25.8	34.7
PionIn3Planes	1 390	2 299	124	71	3 885	24.1	35.8
ShowerCut	1 154	1 097	88	59	2 398	20.0	48.1
OpeningAngle	1 146	1 002	33	18	2 198	19.8	52.1
Nonprotons	**1 064**	**783**	**24**	**9.5**	**1 881**	**18.4**	**56.6**
Combined FHC+RHC (1.885×10^{21} *POT)*							
None	10 530	521 821	6 817 663	437 617	7 787 632	100.0	0.1
InFiducialVol	8 783	118 309	261 981	22 645	411 718	83.4	2.1
Topological	6 738	54 707	9 066	2 071	72 582	64.0	9.3
MuonCandidate	6 052	37 193	3 960	1 259	48 465	57.5	12.5
ContainedPion	2 801	4 621	374	220	8 016	26.6	34.9
MuonIn3Planes	2 684	4 365	311	167	7 527	25.5	35.7
PionIn3Planes	2 500	3 941	241	109	6 791	23.7	36.8
ShowerCut	2 091	1 947	174	91	4 303	19.9	48.6
OpeningAngle	2 077	1 776	66	28	3 947	19.7	52.6
Nonprotons	**1 936**	**1 404**	**48**	**16**	**3 404**	**18.4**	**56.9**

Phase space and opening-angle requirement. The thresholds were chosen from the efficiency turn-on and purity studies of the Run-1 FHC sample (Sec. 21.1 of the technical supplement); the measured phase space excludes the momentum regions of near-zero efficiency and the low-purity region of large opening angle. The events removed by the requirement $\theta_{\mu\pi} < 2.6$ rad form the cosmic control region (Table 13): in FHC 5.4 signal, 62.0 neutrino-background, 42.3 beam-off and 19.2 dirt events, a signal purity of 4.2%. Without the requirement the purity of the selected sample would fall from 57.3% to 53.1% and its beam-off content would rise from 24 to 66 events. No lower bound on the opening angle is applied, since the small-angle region has a purity of 58% and a cut there would remove signal without improving the purity.

9.1 N−1 distributions

Figure 5 shows the N−1 distributions (all cuts applied except the one shown) of the topological score (Cut 3) and the muon–pion opening angle (Cut 9), the two cuts instrumented event by event, for the three configurations. The shapes agree between the horn modes, as expected for an identical selection; the RHC and combined samples are larger. The remaining cut variables are shown at the final cut in §9.3.



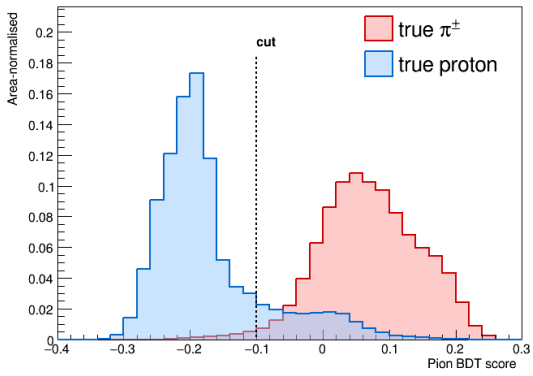
(a) Topo. score, FHC. (b) Topo. score, RHC. (c) Topo. score, comb.

(d) Opening angle, FHC. (e) Opening angle, RHC. (f) Opening angle, comb.

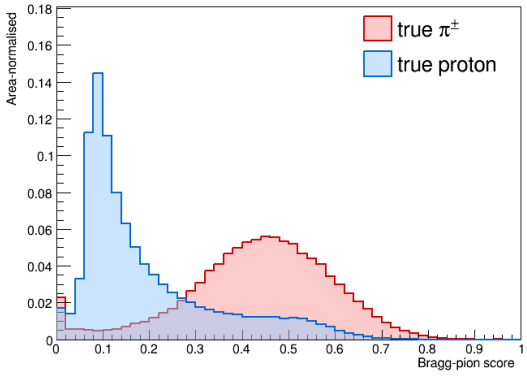
Figure 5: N−1 distributions for the topological score (Cut 3, top) and μ–π opening angle (Cut 9, bottom), for FHC, RHC and combined. Signal (blue) and background (orange) stacked; dashed line = applied threshold. Colourblind-safe Okabe-Ito palette.

9.2 Proton rejection for the pion candidate

The pion BDT provides the proton rejection of the pion identification, together with its LLR and MIP-BDT requirements (Table 10). Figure 6 shows the pion-BDT and Bragg-pion scores of the tracks the identification acts on (end contained, start within 4 cm of the vertex, longer than 20 cm, in events with a muon candidate) for simulated pions and protons. A pion-BDT score above -0.10 keeps 98% of the pions and 16% of the protons. The Bragg-pion score is not used: the beam-on, beam-off and dirt ntuples of FHC Run 1 and RHC Runs 1 and 3 do not carry it, so a requirement on it would apply to the simulation and to only part of the data.



(a) Pion-BDT score; threshold > -0.10.

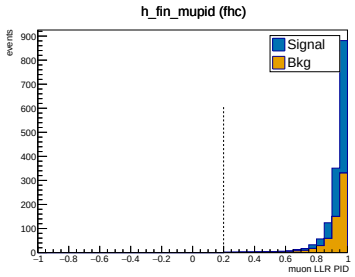


(b) Bragg-pion score (not used).

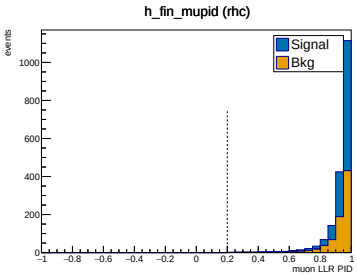
Figure 6: Pion-candidate PID distributions for true $\pi^{\pm}$ (red) and true protons (blue), for the tracks the pion identification acts on, all simulated samples with central-value weights, area-normalised.

9.3 Distributions at the final cut

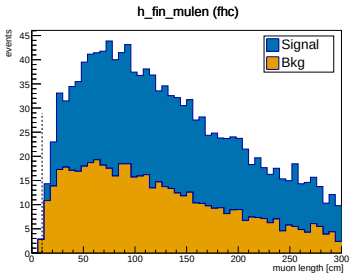
Figures 7–9 show the particle-identification and track-length variables of the two candidates and two topological variables at the final cut, for FHC and RHC.



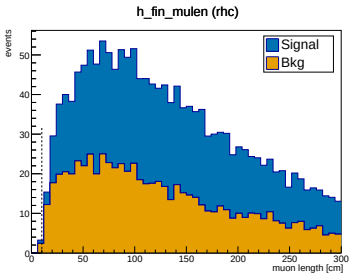
(a) Muon LLR PID, FHC.



(b) Muon LLR PID, RHC.

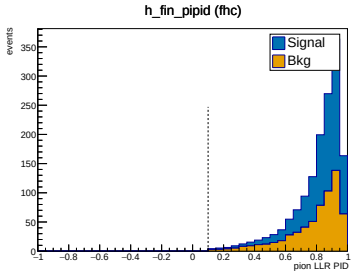


(c) Muon length, FHC.

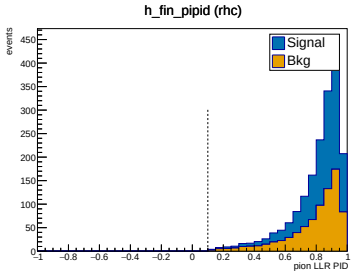


(d) Muon length, RHC.

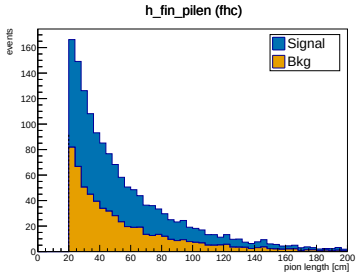
Figure 7: Muon-candidate LLR PID (top) and track length (bottom) at the final cut, for FHC and RHC (the combined configuration is consistent with both and is not shown). Signal (blue) / background (orange) stacked.



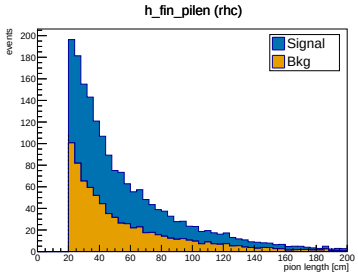
(a) Pion LLR PID, FHC.



(b) Pion LLR PID, RHC.

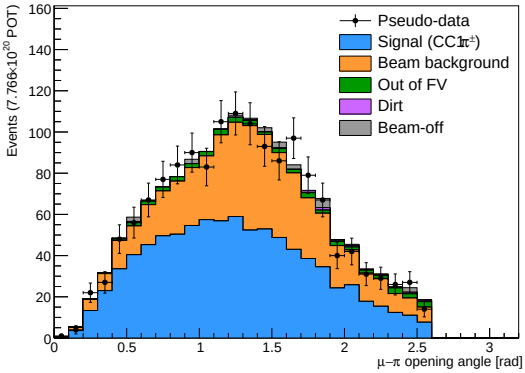


(c) Pion length, FHC.

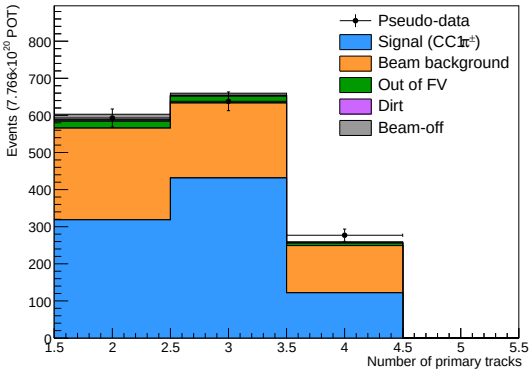


(d) Pion length, RHC.

Figure 8: Pion-candidate LLR PID (top) and track length (bottom) at the final cut, for FHC and RHC (the combined configuration is consistent with both and is not shown). Signal (blue) / background (orange) stacked.



(a) μ–π opening angle.



(b) Number of primary tracks.

Figure 9: Topological variables at the final cut in FHC (7.766×10^{20} POT): the per-run pseudo-data against the simulation, normalised as in the extraction.

9.4 Background composition

Table 12 gives the composition of the background at the final cut.

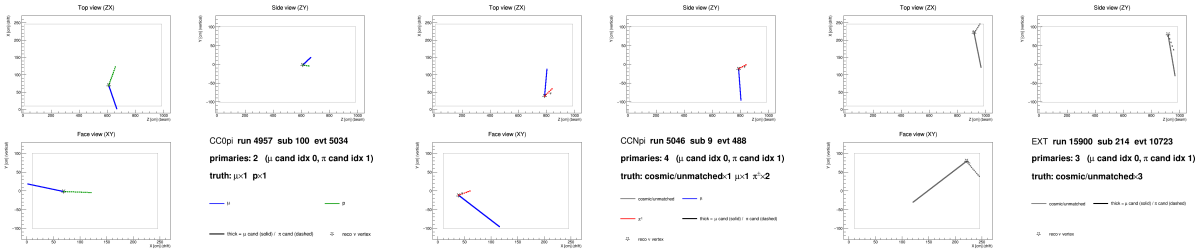
Table 12: Background composition at the final cut, FHC full exposure: truth classes of the selection's counters, weighted with the central-value weight and scaled per run to the recorded exposure (the signal-region column of Table 26 of the technical supplement). Fractions are of the 620 neutrino-background events; the beam-off and dirt yields are listed for scale.

Category	Events	Fraction
ν_μCC 0π (proton or other track as pion)	260	42%
ν_μCC $\geq 2\pi^\pm$, no π^0	131	21%
NC and ν_eCC	82.5	13%
ν_μCC with a π^0	52.7	8%
ν_μCC $1\pi^\pm$ outside the measured phase space	49.6	8%
Out of fiducial volume	44.0	7%
EXT (cosmic)	24	n/a
Dirt	6.6	n/a

The largest background, 42% of the neutrino background, is CC0π interactions in which a proton is reconstructed as the pion candidate. Event displays labelled with the simulated particle type confirm this: the pion-candidate track is a stopping proton with a Bragg peak, while the muon is identified correctly. The pion candidate is a proton in about 21% of the selected events.

9.5 Event displays

Figure 10 shows two-dimensional event displays, labelled with the simulated particle type, of the two largest backgrounds and of a beam-off event. Tracks are coloured by the matched particle: μ (blue), $\pi^\pm$ (red), p (green), cosmic or unmatched (grey). The muon candidate is drawn thick and solid, the pion candidate thick and dashed, and the reconstructed vertex as a star.



(a) CC0π background: a proton (green, dashed) is reconstructed as the pion candidate.

(b) CC$N\pi$ background: two genuine charged pions present; one is selected as the pion candidate.

(c) EXT cosmic event: all tracks are grey (no truth match), faking a neutrino vertex topology.

Figure 10: Truth-labelled 2D event displays (ZX projection) for representative background events at the final cut. Three views (ZX top, ZY side, XY face) are generated per event.

10 Background control regions

The background prediction is tested in four control regions, each defined by inverting one requirement of the signal selection (Fig. 11). No event can belong to a control region and to the signal region. The control regions can be compared with beam-on data while the signal region remains blind. However, the control samples contain signal events not selected by the analysis, so we will look at them after this analysis is approved and before we unblind the signal region.

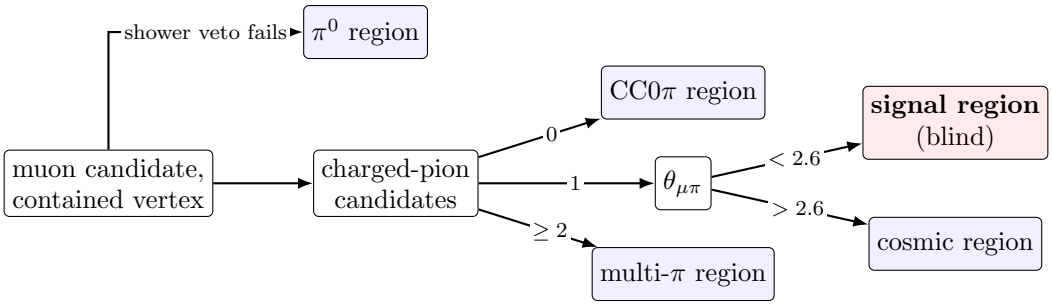


Figure 11: The four control regions and the signal region. Each region inverts one requirement of the signal selection, so no event belongs to a control region and to the signal region; this allows the control regions to be compared with beam-on data while the signal region remains blind.

10.1 Region definitions and expected yields

Each region inverts one requirement of the signal selection; requirements that the inversion leaves undefined are dropped explicitly (branch-level definitions in Table 25 of the technical supplement):

- **CC0π**: the muon selection without a reconstructed charged pion, enriched in quasi-elastic and 2p2h interactions;
- **multi-π**: two or more charged-pion candidates, enriched in resonant and deep-inelastic multi-pion production;
- π^0: a reconstructed shower that fails the shower veto, enriched in charged- and neutral-current π^0 production;
- **cosmic**: the full signal selection with the opening-angle requirement inverted ($\theta_{\mu\pi} > 2.6$ rad), enriched in cosmic-ray activity.

The regions use the per-run normalisation and the central-value weights of the measurement (§5). Table 13 gives the expected yields.

Table 13: Control-region yields at full exposure. "Signal" is the signal content of the neutrino simulation, "ν bkg" the remaining neutrino simulation, EXT the beam-off samples scaled by the trigger ratios of their run periods (§5) and dirt the overlay of interactions outside the cryostat; all yields and the pseudo-data apply Eq. (4). The pseudo-data column is a Poisson throw of the neutrino simulation only and is compared with Signal + ν bkg.

Beam	Control region	Pseudo-data	Signal	ν bkg	EXT	Dirt	Sig. frac.
FHC	CC0π	10 487	943	9 629	1 075	342	8.9%
FHC	multi-π	109	29	91	6.0	0.5	24.3%
FHC	π^0	5 789	582	5 265	613	93	10.0%
FHC	cosmic	80	5.4	62	42	19.2	8.0%
RHC	CC0π	12 600	1 150	11 373	1 212	460	9.2%
RHC	multi-π	194	38	133	6.4	2.8	22.2%
RHC	π^0	8 180	780	7 300	678	124	9.6%
RHC	cosmic	85	6.7	78	43	35.2	7.9%
Comb	CC0π	23 087	2 093	21 002	2 287	802	9.1%
Comb	multi-π	303	67	224	12.5	3.3	23.1%
Comb	π^0	13 969	1 362	12 565	1 291	216	9.8%
Comb	cosmic	165	12.1	140	85	54.4	8.0%

The regions differ in size and in the background they probe:

- **CC0π** is the largest region, with about 10.5k events in FHC, 12.6k in RHC and 23k combined. Its neutrino component is dominated by quasi-elastic and $2p2h$ interactions, which reach the signal region through pion absorption or a proton reconstructed as the pion. It probes the final-state interactions that move events between CC$\pi^{\pm}$ and CC0π, the largest migration path into and out of the signal definition (§6).

- π^0 is the second largest, 5.8–8.2k events per mode and 14k combined. It is enriched in charged- and neutral-current π^0 events, whose photon showers the veto (Cut 8) removes, and it probes the veto efficiency together with the π^0 production rate: these are the events that enter the signal region when a photon is missed.

- **multi-π** is small, 109–194 events per mode (303 combined), and has the largest signal content, 22–24%, because a multi-pion event with one pion lost is the background the selection is most exposed to. Its size limits its constraining power.

- **cosmic** has the largest beam-off share: in FHC 42 beam-off against 67 neutrino events, a ratio of 0.6, against 0.1 in the CC0π region. It probes the beam-off normalisation and shape. Its beam-off purity is 33% in FHC and 26% in RHC, below the target purity of the frozen procedure (Sec. 20.5 of the technical supplement), so it can add an uncertainty to the beam-off normalisation but not correct it.

The combined CC0π and π^0 regions have raw statistical precisions of 0.7% and 0.8%. Their constraint on the signal-region background is weaker, because it passes through transfer factors that depend on the model (Sec. 20.3 of the technical supplement): the factors are small (0.03 and 0.02), but the transfer from a CC0π event to a selected CC1π topology carries its own cross-section and detector uncertainties, which limit the constraint more than the sample size does. The cosmic region would constrain the beam-off normalisation only to about 60%: the combined region contains 85 beam-off events among about 291, with 17 events of statistical uncertainty and a 32% systematic uncertainty on its 152 neutrino events ($\pm$49 events); the statistics alone would give 20%. With the beam-off background at 1.4% of the signal region this is sufficient. The multi-π region, 303 combined events, gives about 6%, useful as a cross-check.

Pseudo-data test. The region definitions and the per-run normalisation are tested on pseudo-data, a Poisson throw of the neutrino simulation alone, compared with the neutrino simulation (Table 14). The two large regions agree with unity to 1% in all three configurations. The small regions scatter within their Poisson uncertainties: FHC multi-π at 0.91 on 109 events (10%), FHC cosmic at 1.19 on 80 events (11%, 1.7σ) and RHC multi-π at 1.14 on 194 events (7%). A variant with the beam-off and dirt components added to the pseudo-data tests the complete bookkeeping (Sec. 20.2 of the technical supplement).

Table 14: Pseudo-data test of the control regions. "Signal fraction" is the signal content of the neutrino simulation in each region. $N_{\text{data}}/N_{\nu\text{MC}}$ is the ratio of a Poisson throw of the neutrino simulation to the simulation itself: the two large regions (CC0π, π^0) are within 1% of unity in the three configurations, and the two small regions (multi-π, cosmic) scatter within their Poisson uncertainties (7–11%). The test checks the region definitions and the per-run normalisation, not the background model.

	Signal frac.	$N_{\text{data}}/N_{\nu\text{MC}}$		
Control region	(of νMC)	FHC	RHC	Combined
CC0π	$\sim 9\%$	0.99	1.01	1.00
multi-π	$\sim 23\%$	0.91	1.14	1.04
π^0	$\sim 10\%$	0.99	1.01	1.00
cosmic	$\sim 8\%$	1.19	1.00	1.08

10.2 Validation of the background fit

The background fit is validated on simulation: toy ensembles give pull widths of 0.98–1.09, injected normalisation shifts are recovered, a likelihood-ratio test separates common from beam-specific normalisations, and a Markov-chain sampler reproduces the profile-likelihood intervals. With 10^5 toys the combined Neyman–Pearson data variance leaves a pull-mean offset of 0.3–0.7% on the normalisations in the two-dimensional fits, which the Pearson form removes, and the fits use that form. The studies are described in Sec. 20.8 of the technical supplement.

11 Cross-section extraction

11.1 Extraction chain

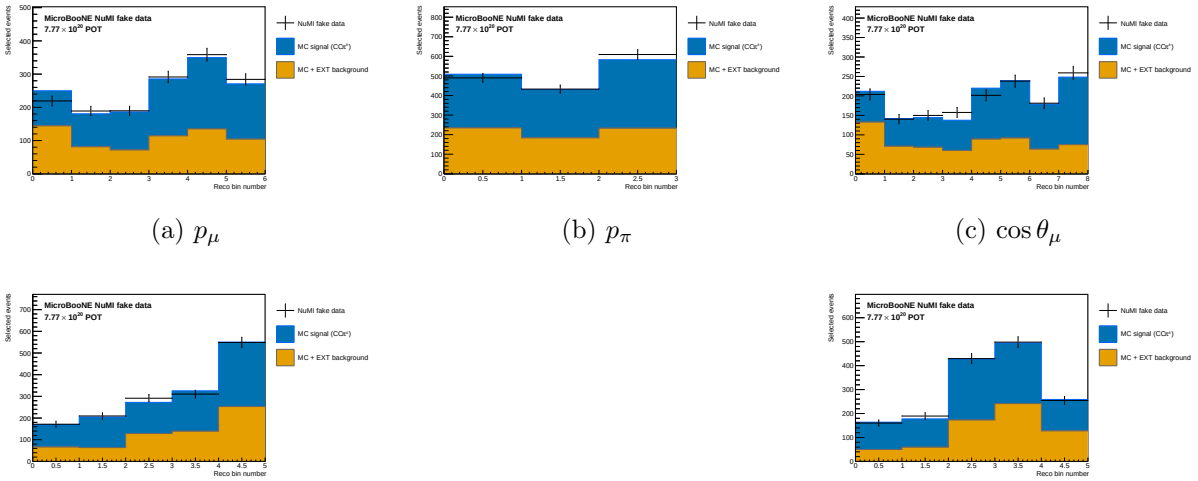
The extraction proceeds in three stages, shown for each configuration in the reconstructed-bin space in which the unfolding operates, with all analysis slices concatenated.

Selected events. The selected reconstructed spectrum is scaled to the exposure of each configuration with the per-run factors of §5, and the simulation is split into signal and background (Fig. 12). At the final cut the purity is 57–59% (Table 11), so about 40% of the selected events are background. The points are the pseudo-data.

Background subtraction. The estimated background is subtracted bin by bin (Fig. 13). The neutrino background is taken from the simulation, the cosmic-ray background from the beam-off data and the background from interactions outside the cryostat from the dirt overlay; the control regions of §10 test this prediction. For observable x the background-subtracted spectrum in reconstructed bin i is

$$d_i^{\text{sig}} = d_i^{\text{data}} - b_i^{\nu\text{-bkg}} - b_i^{\text{EXT}} - b_i^{\text{dirt}}. \tag{9}$$

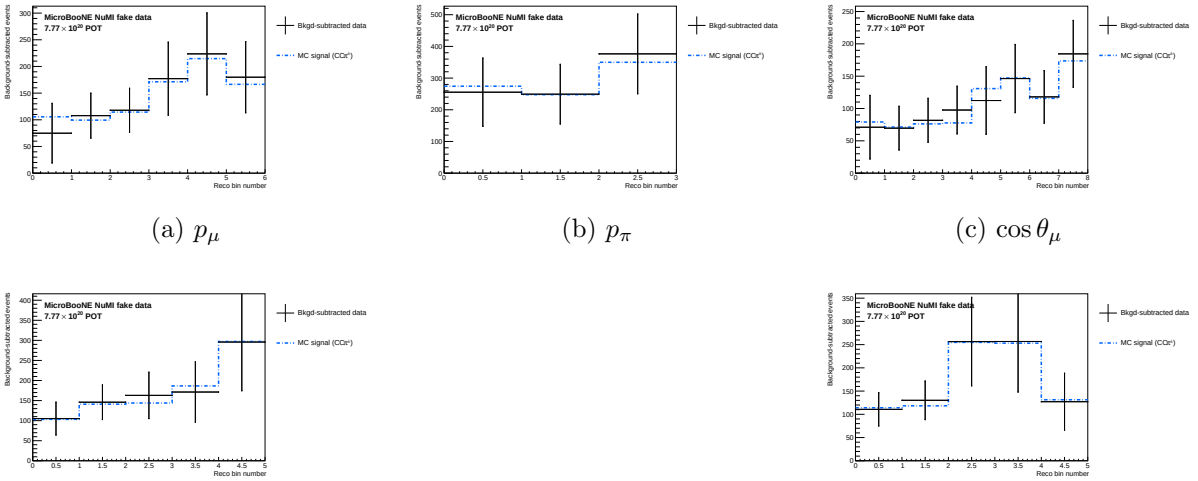
Unfolding. The subtracted spectrum still contains the detector response and the selection efficiency, described by the response matrix of §11.2. With up to half of the selected events of a bin coming from other true bins, a direct inversion is unstable (§11.4), and the spectrum is unfolded with the Wiener-SVD method. The regularisation introduces the additional-smearing matrix A_C: the unfolded result estimates A_C times the true signal, and every prediction is multiplied by A_C before it is compared with the result (§14.4).



(a) p_μ (b) p_π (c) $\cos\theta_\mu$

(d) $\cos\theta_\pi$ (e) $\theta_{\mu\pi}$

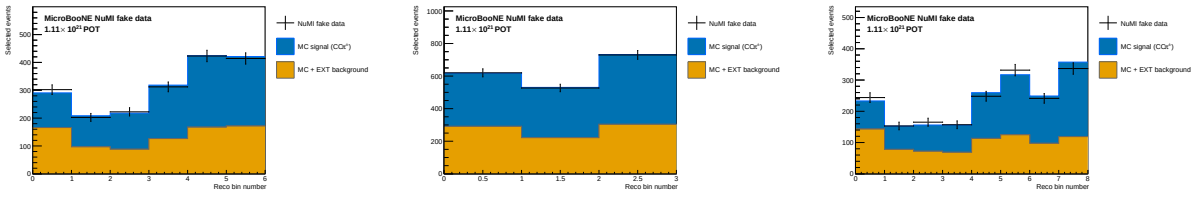
Figure 12: Stage 1, FHC: selected reconstructed spectra scaled to the full exposure, with the MC split into signal and background. Points are the Poisson-thrown pseudo-data. The RHC equivalent is Fig. 14 and the combined one Fig. 9 of the technical supplement.



(a) p_μ (b) p_π (c) $\cos\theta_\mu$

(d) $\cos\theta_\pi$ (e) $\theta_{\mu\pi}$

Figure 13: Stage 2, FHC: background-subtracted data compared with the central-value MC signal; the input to the unfolding. RHC: Fig. 15; combined: Fig. 10 of the technical supplement.

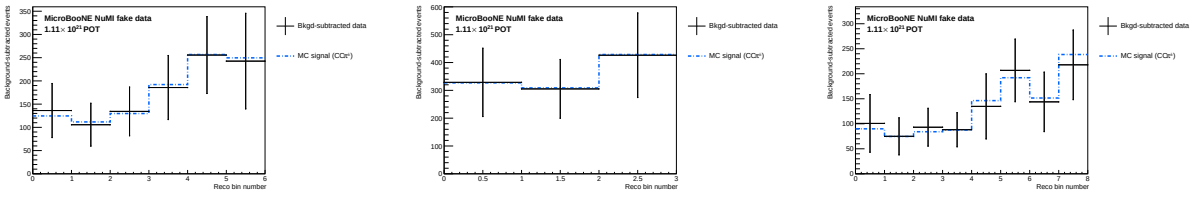


(a) p_μ (b) p_π (c) $\cos\theta_\mu$



(d) $\cos\theta_\pi$ (e) $\theta_{\mu\pi}$

Figure 14: Stage 1, RHC: selected reconstructed spectra scaled to the full RHC exposure, MC split into signal and background.



(a) p_μ (b) p_π (c) $\cos\theta_\mu$



(d) $\cos\theta_\pi$ (e) $\theta_{\mu\pi}$

Figure 15: Stage 2, RHC: background-subtracted data compared with the central-value MC signal.

11.2 Response matrix

Two matrices are used, and they differ by the efficiency. The migration matrix

$$M_{ij} = P(\text{reco in bin } i \,|\, \text{true in bin } j,\ \text{selected}), \qquad \textstyle\sum_i M_{ij} = 1, \tag{10}$$

describes how selected events move between bins. The response matrix also contains the probability to be selected,

$$S_{ij} = \varepsilon_j \, M_{ij}, \qquad \textstyle\sum_i S_{ij} = \varepsilon_j, \tag{11}$$

where ε_j is the ratio of selected to generated signal events in true bin j. The unfolding uses S, built from selected signal events with the same binning on both axes, and the efficiency of Fig. 20 is its column sum. The folding relation that is inverted is

$$n_i^{\text{reco}} = \sum_j S_{ij} \, N_j^{\text{true}} + b_i, \qquad S_{ij} = \varepsilon_j M_{ij}, \tag{12}$$

so the unfolded $\hat{N}_j^{\text{true}}$ is already corrected for the efficiency, and no separate division by ε_j follows (Eq. (13)). Figure 19 shows S as passed to the unfolding, with each column summing to the efficiency of its true bin. The migration diagonals quoted in the text, for example 91%, 51% and 52% for the three p_π regions, are elements of the column-normalised M, obtained by dividing each column of S by its sum.

11.3 Binning and the migration criterion

The migration diagonal of a bin is the element M_{jj} of the column-normalised migration matrix: the fraction of selected signal events generated in true bin j that are reconstructed in bin j. A small diagonal means that the content of the reconstructed bin comes mostly from other true bins and that the unfolded value of the bin depends on the modelled migrations.

The Booster-beam CC1$\pi^\pm$ measurement required every diagonal to exceed 0.68 and every bin to hold at least 100 predicted events. That threshold is a convention of that analysis, not a general requirement, and it was set for a spectrum and estimators different from those here. In this measurement the muon momentum is measured by multiple Coulomb scattering for 63% of the muons, with a 30% resolution comparable to the bin widths, and the range-based pion momentum saturates near 0.34 GeV/c. Under 0.68 the p_μ distribution would be reduced to four bins, p_π to two regions and $\cos\theta_\mu$ to seven bins.

The binnings of this analysis meet a diagonal of 0.50 in FHC and in RHC: at most half of the selected events of a bin come from neighbouring true bins. In addition there are three further requirements (Sec. 2 of the technical supplement). Every bin holds at least 70 selected signal events (50 in the proton-tagged sample), so that the response is not dominated by its own statistical fluctuations. Among the binnings that pass, the one with the fewest bins that reaches 90% of the largest trace of A_C is chosen; the trace counts the degrees of freedom that the data determine, so a bin whose content would be supplied by the regularisation adds nothing to it and is not added. Finally, the stronger dependence on the modelled migrations that the 0.50 criterion admits is tested directly on the adopted binnings by the ensembles and by the pseudo-data from reweighted models (§13).

Three inclusive observables meet 0.50 but not 0.68: p_μ (smallest diagonal 0.57 in FHC, 0.59 in RHC), p_π in three regions (0.51, 0.53) and $\cos\theta_\mu$ (0.66, 0.66). $\cos\theta_\pi$ and $\theta_{\mu\pi}$ meet 0.68. In the proton-tagged sample $\delta\alpha_T$, $\delta\phi_T$ and $\theta_{\pi p}$ meet only 0.50 (Sec. 3 of the proton-tagged note).

11.4 Regularised unfolding

Unfolding estimates the true signal spectrum $\vec{s}$ from the measured $\vec{m}$ through the folding relation $\vec{m} = A\,\vec{s} + \vec{b}$. The direct inverse $A^{-1}(\vec{m} - \vec{b})$ is unbiased, but bin migration makes A nearly singular, and the inverse turns statistical fluctuations into large anticorrelated oscillations between bins. Regularisation reduces this variance at the cost of a controlled bias. Two methods are used.

Wiener-SVD (primary). After a singular-value decomposition of the response, each mode k is weighted by the Wiener filter $W_k = \sigma_k^2/(\sigma_k^2 + N_k)$, the weighting that minimises the mean squared error: it is close to one for well-measured modes and tends to zero for modes dominated by noise, and its strength is not tuned by hand. The resulting linear bias is known: the unfolded data estimate the smeared spectrum $A_C\vec{s}$, and the additional-smearing matrix A_C is provided with every result so that each prediction is smeared in the same way before comparison.

D'Agostini iterative Bayesian [27] (cross-check). Bayes' theorem inverts the migration, $P(C_j|E_i) = R_{ij}\pi_j / \sum_l R_{il}\pi_l$, and the unfolded estimate is $\hat{s}_j = \varepsilon_j^{-1} \sum_i P(C_j|E_i)(m_i - b_i)$. The prior π_j, initially the simulation, is replaced by the result and the procedure iterated; the number of iterations is the regularisation parameter, with few iterations biased towards the prior and many iterations noisy. There is no A_C, and predictions are compared directly. The residual bias

is towards the prior and can only be evaluated with pseudo-data that differ from the prior. The uncertainty is the data covariance propagated through the iterations at fixed prior.

Table 15: Trade-offs of the two unfolding methods. Neither is "more correct"; they make complementary choices about how to trade bias for variance and how to expose the residual bias.

	Wiener-SVD	**D'Agostini (iter. Bayes)**
Regularisation	Wiener filter (automatic, from S/N)	Iteration count n (chosen)
Residual bias	Explicit A_C, published	Implicit; set by early stopping
Result integral	Suppressed, observable-dependent	Approximately preserved in nominal closure; prior-biased in general
Theory comparison	Smear theory by A_C	Raw theory
Positivity	Not guaranteed	Guaranteed for positive input
Uncertainties	Closed-form, linear	Correlated across iters.; needs care

Wiener-SVD provides the results, for its objective regularisation and its explicit bias A_C. D'Agostini provides a cross-check (Sec. 12 of the technical supplement). Its integral is independent of the iteration count and of the observable, and lies 27–30% above the Wiener-SVD integrals averaged over the observables, depending on the configuration; this is consistent with the spread of the Wiener-SVD integrals between observables (Table 23) being caused by the A_C smearing, which the row sums of §14.4 show directly. With pseudo-data equal in expectation to the central-value simulation that is also its prior, D'Agostini returns its prior, so the stability of its integral does not measure its bias for a different truth; that requires pseudo-data from a different model (§13.3).

11.5 Wiener-SVD implementation and normalisation

The Wiener-SVD unfolding [7] of the XSecAnalyzer framework [18, 17] produces every result: the data covariance is whitened by a Cholesky decomposition, the regularisation is applied inside the SVD of RC^{-1}, and the Wiener filter of Eq. 3.24 of Ref. [7] is set by the data covariance. The regularisation matrix C is the second-derivative operator on the density, corrected for the non-uniform bin widths, with one-sided first-derivative rows at the ends. The width correction matters because the bin widths vary by up to a factor of 56 ($\cos\theta_\mu$; 14 for p_μ): a stencil for a uniform grid weights the curvature of the density wrongly and drives A_C towards rank one, so that every model is smeared onto one shape and the result carries no shape information (Sec. 24 of the technical supplement). A standalone replica of the unfolding is used for cross-checks between pipelines; it takes the covariance from the framework because it generates no universes, and with a statistical covariance alone it under-regularises and over-corrects the integral by about 26%, so it is not used for any result.

The differential cross section in true bin j is

$$\left.\frac{d\sigma}{dx}\right|_j = \frac{\hat{N}_j^{\text{true}}}{\Delta x_j \cdot \Phi_{\text{total}} \cdot N_{\text{Ar}}}, \tag{13}$$

where $\hat{N}_j^{\text{true}}$ is the Wiener-SVD estimate of the true signal count obtained by inverting Eq. (12), $\Phi_{\text{total}} = \Phi_{\text{per POT}} \times T_{\text{POT}}$ the integrated flux and N_{Ar} the number of argon nuclei in the FV. Since the unfolding uses $S = \varepsilon M$, the efficiency correction is part of the inversion and no factor $1/\varepsilon_j$ appears. The implementation forms each bin as $\hat{N}_j^{\text{true}}/(\text{conv} \cdot \Delta x_j)$ with the conversion $\text{conv} = N_{\text{Ar}}\,\Phi_{\text{total}}/10^{-38}$, whose inputs are:

Quantity	FHC	RHC	Combined
N_{Ar} (FV, unresponsive region excluded)		8.710×10^{29}	
Flux constant ϕ ($E_\nu > 60$ MeV; 10^{-10} cm^{-2}/POT)	6.81159	6.44646	6.59691
Data POT (10^{20})	7.766	11.082	18.848
Integrated flux $\Phi_{\mathrm{total}} = \phi \times \mathrm{POT}$ (10^{11} cm^{-2})	5.290	7.144	12.434
conv $= N_{\mathrm{Ar}}\,\Phi_{\mathrm{total}}/10^{-38}$ (events per 10^{-38} cm^2/Ar)	4608	6223	10830

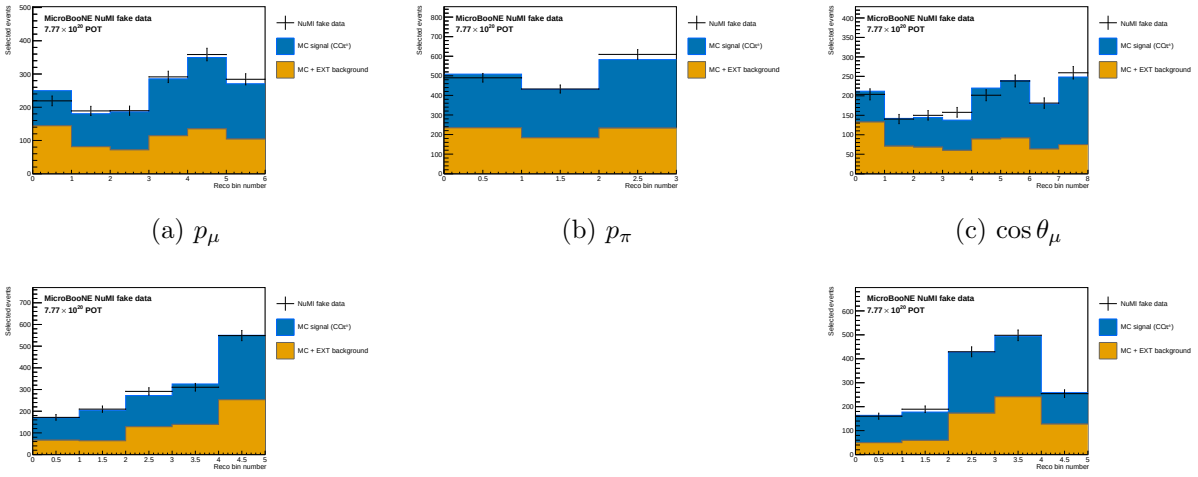
The integrated cross section of a differential result is the width-weighted sum

$$\sigma_{\mathrm{int}} = \sum_j \left.\frac{d\sigma}{dx}\right|_j \Delta x_j. \tag{14}$$

Since the Wiener-SVD result estimates $A_C\, x_{\mathrm{true}}$ and A_C does not preserve the normalisation, σ_{int} differs between observables (§14.4). The total cross section, independent of the observable, is the one-bin extraction of §14.6.

11.6 Reconstructed spectra, response and efficiency

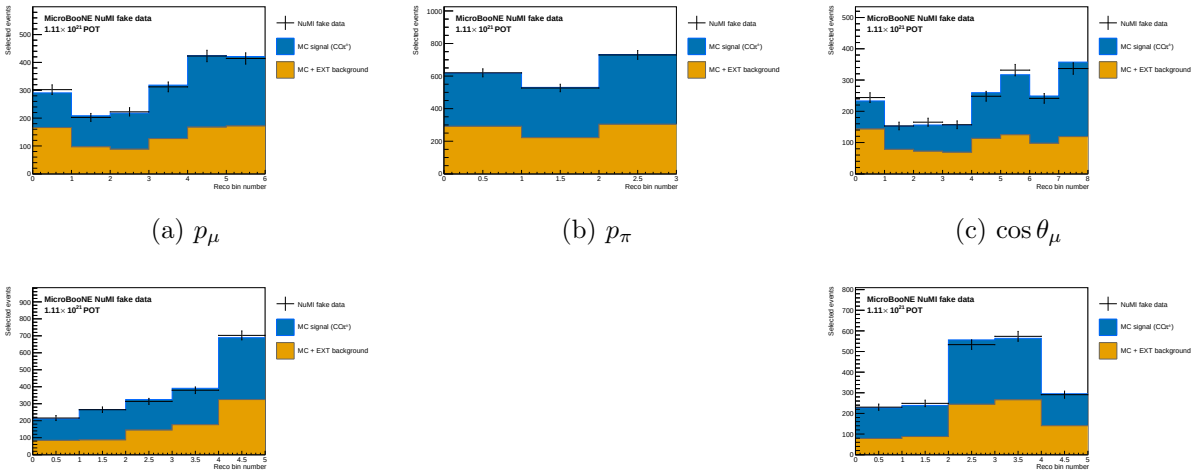
Figures 16–18 show the reconstructed spectra of the five observables at the final cut, with the simulated signal and the background components.



(a) p_μ (b) p_π (c) $\cos\theta_\mu$

(d) $\cos\theta_\pi$ (e) $\theta_{\mu\pi}$

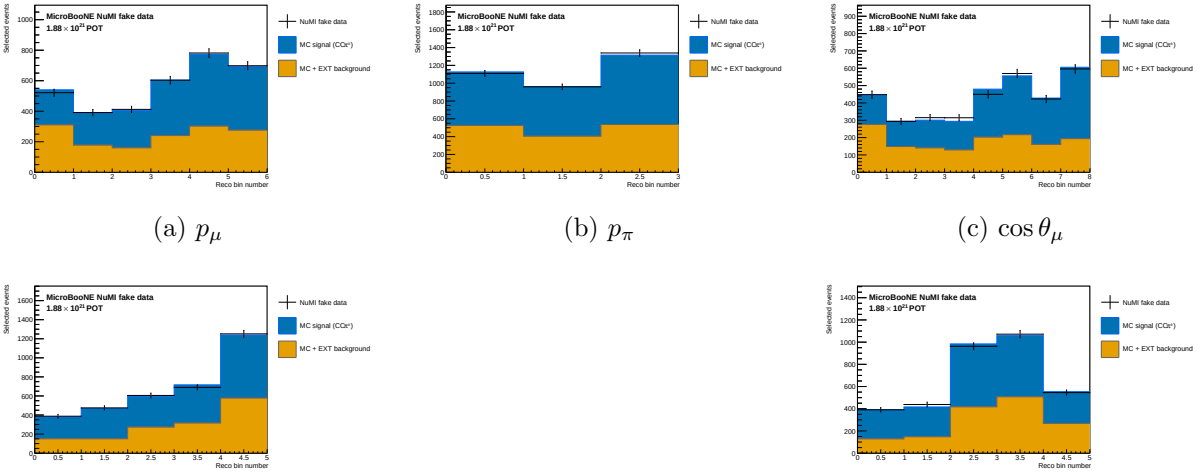
Figure 16: Reconstructed-space spectra at the final cut for all five observables, FHC full exposure (Runs 1+2+4+5): Poisson-thrown central-value pseudo-data (black points), stacked signal (blue), beam background (orange), dirt (green), and EXT (grey). The pseudo-data is drawn from the central-value Monte Carlo and so agrees with the stacked prediction by construction (a reco-level self-closure); the differential results and their closure against the A_C-smeared truth are shown in Fig. 23.



(a) p_μ (b) p_π (c) $\cos\theta_\mu$

(d) $\cos\theta_\pi$ (e) $\theta_{\mu\pi}$

Figure 17: Reconstructed-space spectra, RHC full exposure (Runs 1+2+3+4a+4b+4c); same components as Fig. 16.



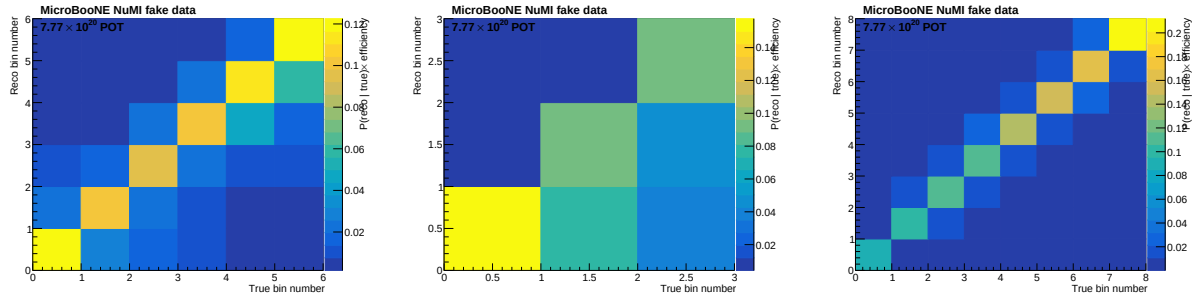
(a) p_μ (b) p_π (c) $\cos\theta_\mu$

(d) $\cos\theta_\pi$ (e) $\theta_{\mu\pi}$

Figure 18: Reconstructed-space spectra, combined FHC+RHC (per-mode normalised); same components as Fig. 16.

Figure 19 shows the response matrices of the five observables. The column-normalised diagonals of the FHC responses are 0.86–0.91 in the best-resolved bin of each angular observable (the forward bin of $\cos\theta_\mu$ and the first bin of the others) and 0.66–0.88 elsewhere in $\cos\theta_\mu$, $\cos\theta_\pi$ and $\theta_{\mu\pi}$. For the six p_μ bins they are 0.75, 0.64, 0.58, 0.59, 0.58 and 0.57: the MCS estimator, which measures 63% of the selected muons, has a resolution of about 30%, comparable to the widths of the p_μ bins. Every binning meets the 0.50 criterion in FHC and in RHC (App. A), and the RHC diagonals differ from the FHC ones by at most 0.04.

At any fine binning the p_π response is far from diagonal, because the range estimator saturates near 0.34 GeV/c once pions interact before stopping; the fine-bin study is in Sec. 6.2 of the technical supplement. The three-region response has column-normalised diagonals of 91%, 51% and 52%. It separates the two intervals near threshold from the region above them; the upper region is an inclusive region rather than a resolved point of a spectrum (§14.3).



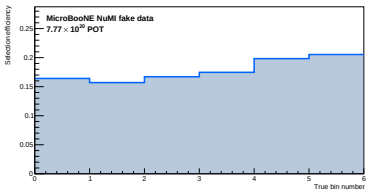
(a) p_μ (b) p_π (c) $\cos\theta_\mu$



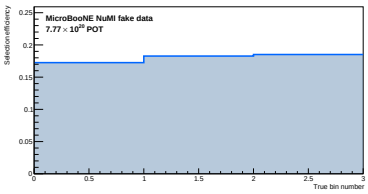
(d) $\cos\theta_\pi$ (e) $\theta_{\mu\pi}$

Figure 19: Response matrices $S_{ij} = \varepsilon_j M_{ij}$, FHC, exactly as passed to the unfolder, at the binning of the results, including the three-region p_π response (91%, 51% and 52% column-normalised diagonals); the rejected five-bin p_π response, which shows the reconstructed events piling into the lowest bin from every true bin, is Fig. 11 of the technical supplement.

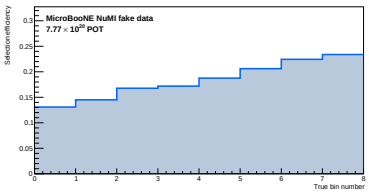
Figure 20 shows the selection efficiency in each true bin, the ratio of selected to generated signal events.



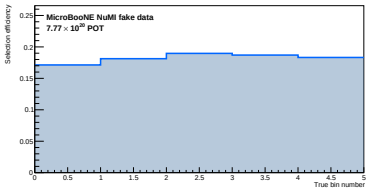
(a) p_μ



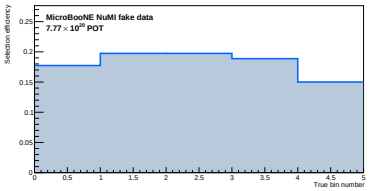
(b) p_π



(c) $\cos\theta_\mu$



(d) $\cos\theta_\pi$

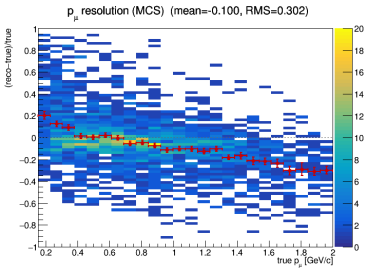


(e) $\theta_{\mu\pi}$

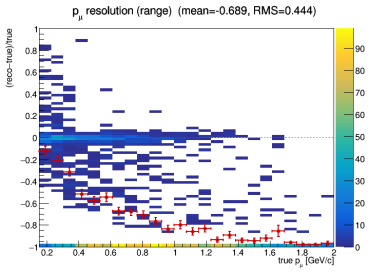
Figure 20: Selection efficiency as a function of true observable value, in the measured phase space ($p_\mu > 0.15$, $p_\pi > 0.175\,\mathrm{GeV}/c$, $\theta_{\mu\pi} < 2.6$ rad). Efficiency is the ratio of selected to generated signal MC ($\varepsilon_j = \sum_i S_{ij}$, summed over reconstructed bins). With the momentum thresholds and the 2.6 rad opening-angle cut the efficiency is flat at 13–20% across all observables; the low-momentum turn-on and the large-angle region dominated by cosmic rays lie outside the measured phase space. The lowest point is the smallest opening-angle bin (9% efficiency), which is kept because its purity is high (58%).

11.7 Reconstruction resolution

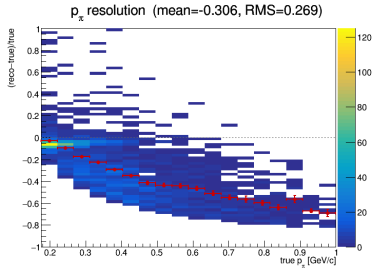
Figure 21 shows the resolution of each observable for selected signal events. For the momenta the relative residual (reco − true)/true is shown, for the angles the absolute residual (reco − true), since the relative residual diverges as $\cos\theta \to 0$ or $\theta_{\mu\pi} \to 0$. The red points are the mean residual in each true bin; the quoted numbers are the overall mean (bias) and RMS (resolution).



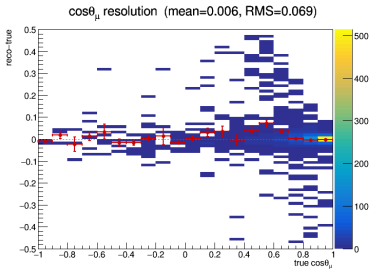
(a) p_μ (MCS): bias −10%, RMS 30%.



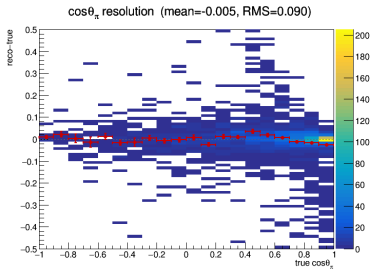
(b) p_μ (range): valid only for contained muons.



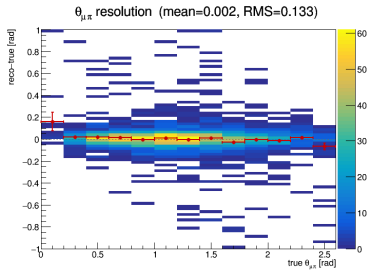
(c) p_π: bias −31%, RMS 27% (analysis estimator).



(d) $\cos\theta_\mu$: RMS 0.069.



(e) $\cos\theta_\pi$: RMS 0.090.



(f) $\theta_{\mu\pi}$: RMS 0.133 rad.

Figure 21: Reconstruction resolution vs. true value, selected signal. Momenta: relative (reco − true)/true; angles: absolute (reco−true). Red points: mean residual per true bin. The muon MCS momentum slightly over-estimates at low p_μ and increasingly under-estimates above ∼1 GeV/c; the range estimator is badly biased for *exiting* muons (range = distance to wall), which is why the analysis uses range only for contained muons (range-or-MCS). The pion panel shows the estimator the analysis actually bins on, not the raw muon-hypothesis range momentum (Sec. 6.2 of the technical supplement): signed mean bias $b = -31\%$, RMS 27%. That average is dominated by high momenta; b is only −2% in the first true bin and reaches −65% above 0.8 GeV/c, because the reconstructed value saturates near 0.34 GeV/c once pions interact hadronically before stopping. The angular variables are largely unbiased with narrow resolution.

12 Systematic uncertainties

Every result carries flux, interaction-model, hadron re-interaction, detector, muon-momentum-scale, POT, target-count and statistical terms. Each is a covariance matrix built from the universes or variation samples of its source, propagated through the unfolding with the error-propagation matrix and provided per component (§14.4); Table 16 lists the sources and their sizes. The detector term applies the fractional change of each variation relative to the detector-variation CV sample to the nominal prediction, since the variation samples were simulated with the earlier NuMI flux and do not share the event rate of the nominal overlays (§14). The muon-momentum-scale term is evaluated on the data side outside the extraction framework (§12.3) and added to the framework covariance. Three identified effects lie outside the official covariance: the beamline-geometry flux term, the beam-off trigger ratios and the Run-2 beam-off substitution (§12.4).

Table 16: Systematic sources and their bin-averaged fractional contribution, computed from the per-component covariance matrices of the fifteen inclusive differential results. Ranges span observables and configurations.

Source	Branch / method	Range (%)
Flux (PPFX multisims)	`weightsFlux`	24.1–36.5
Detector response	Dedicated samples	12.1–24.9
Cross-section model	`weightsGenie`	9.5–15.8
Hadron re-interaction	`weightsReint`	3.5–12.8
POT counting	Beam toroids	2.9–4.2
Target count	FV geometry	1.4–2.1
MCS momentum scale	data-side variation	0.0–13.4
MC statistics	universe spread	2.1–5.4
EXT statistics	beam-off sample	0.6–1.8
Data statistics	thrown pseudo-data	7.1–17.0
Prediction total	quadrature sum	31.1–49.4
Total	incl. data stats	32.5–52.1

Flux and detector response are the largest terms; which of the two dominates depends on the observable and configuration (Table 17; Table 21 of the technical supplement–Table 22 of the technical supplement).

12.1 Breakdown by source

Table 17 gives the bin-averaged fractional uncertainty of each source group for the FHC extractions; the RHC and combined tables, from the same dumps, are Table 21 of the technical supplement and Table 22 of the technical supplement, and the values agree with Table 20.

Table 17: FHC (Runs 1, 2, 4, 5): bin-averaged fractional uncertainty (%) by source group for the five extractions, from the systematics dumps of the extraction. "Prediction total" excludes the data statistics; the flux row is the PPFX multisim term. Every row is a bin average of the per-bin fractional uncertainties, so the rows do not combine in quadrature to the totals (the quadrature is taken per bin before averaging).

Source	p_μ	p_π	$\cos\theta_\mu$	$\cos\theta_\pi$	$\theta_{\mu\pi}$
Prediction total	**37.0**	**49.4**	**36.7**	**35.8**	**33.5**
Cross section (GENIE)	11.0	15.8	11.2	11.3	10.1
Flux (PPFX)	28.6	36.5	28.0	27.1	28.3
Detector	16.1	24.9	18.8	17.6	13.5
Reinteraction	4.1	12.8	3.5	5.1	3.8
MCS momentum scale	10.1	0.0	0.0	0.0	0.0
MC stat	3.4	5.2	5.4	4.4	2.5
EXT stat	0.7	1.8	1.6	1.2	0.8
Data stat	10.9	16.4	17.0	14.1	7.9
POT + targets	3.6	4.7	3.6	3.4	3.5
Total (incl. data stat)	**38.6**	**52.1**	**40.7**	**38.6**	**34.5**

The NuMI flux uncertainty has a hadron-production part, the PPFX multisim term that dominates the budget, and a beamline-geometry part. The latter is built from the twelve geometry variations stored as weights in the overlay (horn current, horn positions, beam spot size and position, horn water layer and target position), each treated as a two-sided unisim and summed in quadrature.

Evaluated separately it is 1.6–3.3% across the observables. It is not part of the covariance, whose configuration omits the beamline block; added in quadrature to the 28% PPFX term it would raise the flux row by at most 0.2% absolute. Figure 22 shows the breakdown for the three configurations.

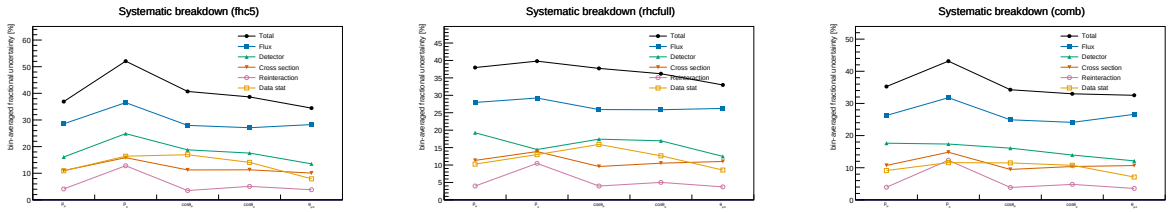


Figure 22: Bin-averaged fractional systematic uncertainty by source vs. observable, for FHC (left), RHC (centre) and combined (right); the figure form of Table 20 of the technical supplement–Table 22 of the technical supplement: total (black), flux (blue), detector (green), cross section (vermillion), reinteraction (purple), data stat (orange).

12.2 Flux uncertainty and background subtraction

The NuMI flux normalisation is uncertain by about 17%, while the flux term of the one-bin total cross section is 28–30% (Table 27) and that of the differential results 24–29% (Tables 20–22; 29–37% for the three-region p_π). The difference comes from the background subtraction, not from the unfolding: the one-bin total has no regularisation and carries the same factor. A flux universe scales the whole reconstructed prediction, signal and neutrino background together, and the covariance is that of the measured quantity $(N_{\mathrm{pred}} - B)/(\varepsilon\Phi N_{\mathrm{Ar}})$, with B and Φ moving together in every universe. Relative to the signal, the flux variation is therefore

$$\frac{\delta\sigma}{\sigma} = \frac{\delta\Phi}{\Phi}\left(1 + \frac{B}{S}\right), \tag{15}$$

with B the neutrino background; the beam-off and dirt contributions carry no flux weight. For FHC, $B/S = 620/872$ (Table 11) and a reconstructed-level flux term of 17.7% give 30.3%, against 30.4% for the one-bin extraction. For RHC, $15\% \times (1 + 783/1064) = 26.0\%$ against 28.4%. The differential flux terms show the same effect bin by bin; the three-region p_π, whose response has the lowest diagonals, carries the largest (37% in FHC).

The flux term is thus fully correlated between the normalisation and the subtracted background, which is correct when both come from the same flux prediction: 17% is the normalisation part, and the remainder is the flux dependence of the background model. It can be reduced by a higher purity or by a background constraint from the control regions (§10), not by the regularisation. The reconstructed-level flux term is 16.4–17.7% bin-averaged and flat across observables, as expected of a normalisation term; the 600 PPFX universes reweight the same events, so the term carries no simulation-statistics component.

12.3 Muon momentum scale

The momentum of 63% of the selected muons is measured by multiple Coulomb scattering, whose scale and resolution are not varied by the detector-variation samples. The momentum of a selected muon is exactly the range value when the track is contained and exactly the MCS value when it exits, so the MCS component can be scaled on its own. A momentum-scale error is a difference between data and simulation, so the variation is applied to the data side: the pseudo-data are thrown again with every MCS-measured muon momentum multiplied by 1 ± 0.05, while the simulation, the response and the selection stay nominal, and each variation is unfolded with the nominal response. The half-difference of the two results is taken as a fully correlated

term. The size of the variation, ±5%, is assumed; it has not been compared with the difference between the MCS and range momenta of contained muons in data.

For FHC $d\sigma/dp_\mu$ the half-difference is 7–10% of the bin value in the five lowest bins, 0.17–0.26σ of the per-bin uncertainty, and 19% with the opposite sign in the open highest bin, 0.65σ: an upward scale moves exiting muons into that bin, whose spectrum falls steeply. The integral moves by −1.7% and −0.3% for the upward and downward scale. The combined configuration, in which FHC and RHC are scaled together as one common MCS scale, behaves like FHC: 8–10% in the five lowest bins and 16% in the highest, at most 0.52σ, with the integral moving by −0.4% and +1.5%. In RHC the five lowest bins move by 11–16% (0.29–0.42σ) and the highest by 16% (0.37σ); the RHC integral moves by −6.1% and +2.3%. Adding the term raises the per-bin total uncertainty from 29–41% to 34–42% (FHC, with 29% to 34% in the highest bin), from 31–38% to 35–38% (combined) and from 37–41% to 39–44% (RHC). The closure χ^2 and the biases of the robustness tests (§13) are quoted with the framework covariance, since they test the extraction rather than the uncertainty of the data.

The selection applies no requirement on the reconstructed muon momentum, and no other inclusive observable depends on it. The scaled pseudo-data therefore give results identical to the nominal ones for $\cos\theta_\mu$, $\cos\theta_\pi$, $\theta_{\mu\pi}$, p_π and the total cross section, as verified by extraction for FHC $\cos\theta_\mu$, $\cos\theta_\pi$, p_π and the total at the earlier binnings. The reconstructed bins of these observables do not use the muon momentum, so this holds at any binning, and the term is zero for these results. The comparison of the control regions with data is also unaffected, since it is made at reconstruction level with the same estimator in data and simulation.

12.4 Effects not included in the covariance

Three effects are evaluated or bounded but not included in the official covariance. The beamline-geometry flux term (1.6–3.3%) is discussed above. The covariance contains the statistical uncertainty of the beam-off samples but no uncertainty on the per-run trigger ratios, nor on the use of the Run-1 beam-off sample for Run 2, which is 16% of the FHC and 23% of the RHC exposure. With the beam-off background at 1.6% of the selected sample, a 20% uncertainty on either would add 0.3% to the total uncertainty.

13 Validation

Closure under the nominal model tests the implementation: the matrices, the binning, the normalisation and the bookkeeping. A result can close and still be determined largely by the modelled migrations or by the regularisation, which the migration criterion and the trace of A_C test separately (§11.3). Every inclusive extraction closes, with $\chi^2 \leq 3.12/8$ and ratios of unfolded to true integral of 0.959–1.035.

Ensembles of pseudo-experiments test whether the statistical uncertainties are correctly sized. Six one-dimensional extractions and the two-dimensional result were each repeated on 100 pseudo-data sets; the pull widths are 0.92–1.04 and the 68% coverage is 64–72% (§13.2).

Pseudo-data from reweighted interaction models measure the bias from the regularisation and the nominal response when the true model differs from the nominal one. Two variations were unfolded in FHC and RHC; the largest bias is 1.21σ, in one sparse $\cos\theta_\mu$ bin (§13.3). Reweighting reaches only changes of the parameters of the nominal model, and no closure test with a generator that has its own detector simulation has been made. The control regions, which test the background model, have been studied with pseudo-data only (§10).

13.1 Closure under the nominal model

Unfolding a spectrum thrown from the simulation that built the response detects an inverted matrix, a misaligned binning or a normalisation applied twice. It is reported with every extraction

as the χ^2 against the A_C-smeared truth of the throw, and the ratio of unfolded to true integral averages 1.001 over the fifteen inclusive extractions (0.959–1.035). The test cannot detect a wrong model, since the model is the same on both sides, nor a wrongly sized uncertainty, since a single throw does not sample the distribution of throws. The closure χ^2 uses the full covariance, whose systematic terms are common to the pseudo-data and the reference and cancel in the residual while still entering the covariance; its p-values therefore cluster near unity and are not a sensitive test. The statistical coverage is tested with ensembles.

13.2 Ensemble coverage

One hundred independent Poisson realisations of the data of each configuration were each carried through the whole chain, with the universes rebuilt from that realisation, and the pull in each bin, $p_i = (\hat{x}_i - (A_C x_{\mathrm{CV}})_i)/\sigma_i$, was formed against the central-value truth smeared by the A_C of that realisation. Six extractions were chosen to span the conditioning and the configurations: p_μ and $\cos\theta_\mu$ in FHC (the A_C row sums of the latter fall to 0.50), $\cos\theta_\mu$ in RHC (0.44), the three-region p_π in the combined configuration, and the one-bin totals of FHC and RHC, which have no regularisation. All six use the binnings of the results.

Reference: statistical covariance	pull mean	pull width	68% / 95% coverage
$d\sigma/dp_\mu$, FHC	−0.10	1.03	67.3% / 94.5%
$d\sigma/d\cos\theta_\mu$, FHC	−0.05	1.00	69.8% / 95.0%
$d\sigma/d\cos\theta_\mu$, RHC	−0.05	1.04	63.5% / 93.5%
$d\sigma/dp_\pi$ (three regions), combined	−0.08	0.94	72.3% / 95.3%
one-bin total, FHC	−0.04	0.92	71.0% / 96.0%
one-bin total, RHC	−0.16	1.01	66.0% / 95.0%
expected	0	1	68% / 95%

With 100 members each width is determined to about ±13%. The widths are consistent with one and the coverage with its nominal value, so the statistical uncertainties are correctly sized in the three configurations, and the strongly suppressed observables behave like the well-conditioned ones. Against the full framework covariance, without the MCS term, the pull widths are 0.25–0.44 (0.11–0.12 for the totals) with nearly complete coverage, because the ensemble varies only the data statistics while the full uncertainty also contains the flux, detector and cross-section terms, which are the same in every member; this is the origin of the closure p-values near unity. The ensemble means lie below the reference by 0.1–0.9%, at 0.3–1.5 standard deviations of the mean, and the pull means are −0.04 to −0.10 (−0.16 for the RHC total). This common negative offset of under 1% is well inside the statistical uncertainty of a single measurement and is not corrected. The two-dimensional result has its own ensemble (§14.7), and the proton-tagged extractions have theirs (Sec. 5.2 of the proton-tagged note). The construction of the reference and the full-covariance pulls are described in Sec. 17 of the technical supplement.

13.3 Robustness to reweighted interaction-model variations

Here the pseudo-data are thrown from a reweighted interaction model, while the response, the efficiency and the background prediction remain nominal, so the bias that appears is the combined effect of the regularisation and of a response built from the wrong model. The pseudo-data universe of the framework is filled from the thrown files at truth and reconstruction level, so the truth used for comparison is that of the alternative model, and the reweighting applies to signal and background alike. The pseudo-data are reconstructed GENIE events reweighted event by event to an alternative parameter set, with the same detector simulation, unfolded with the nominal response and compared using the nominal covariance. This is a model-distortion test; a test with a different generator and its own detector simulation would be stronger and has not been made, as the available NuWro sample did not follow the NuMI flux distribution.

Two variations are used, in FHC and RHC. The first is GENIE multisim universe 545, a simultaneous variation of all interaction parameters, chosen because it changes the shapes more than the rate: the signal rate changes by 3.5% in FHC and 2.5% in RHC, while the true distributions in the released bins change by up to 30% in $\cos\theta_\pi$ and 22% in p_π. The second is the $\Delta \to N\pi$ decay angular unisim, a targeted change of the pion angular distribution in resonance events. Both changes are larger than the differences between the tune and the standalone generators.

Table 18: Unfolding pseudo-data from an alternative interaction model with the nominal response, FHC (upper block) and RHC (lower). "Ratio" is the mean of unfolded over the alternative model's own truth, over the bins holding at least 10% of the largest truth; "max dev" is the largest per-bin departure from unity among those bins, and "bias" the largest per-bin departure over every bin, in units of the framework uncertainty of that bin (without the MCS term).

Observable	Alternative model	mean ratio	max dev.	largest bias
p_μ	GENIE multisim u545	0.941	16.2%	-0.48σ
	$\Delta \to N\pi$ angular	1.045	27.7%	$+0.77\sigma$
$\cos\theta_\mu$	GENIE multisim u545	0.896	53.8%	-1.21σ
	$\Delta \to N\pi$ angular	0.902	43.7%	-1.02σ
$\cos\theta_\pi$	GENIE multisim u545	0.993	24.0%	$+0.67\sigma$
	$\Delta \to N\pi$ angular	0.988	18.2%	$+0.47\sigma$
p_π (three regions)	GENIE multisim u545	0.652	57.0%	-0.95σ
	$\Delta \to N\pi$ angular	1.046	8.7%	$+0.19\sigma$
RHC				
p_μ	GENIE multisim u545	0.817	32.4%	-0.90σ
	$\Delta \to N\pi$ angular	0.992	8.9%	$+0.25\sigma$
$\cos\theta_\mu$	GENIE multisim u545	0.992	18.3%	$+0.43\sigma$
	$\Delta \to N\pi$ angular	1.014	6.7%	-0.50σ
$\cos\theta_\pi$	GENIE multisim u545	0.860	22.0%	-0.60σ
	$\Delta \to N\pi$ angular	0.941	11.9%	-0.33σ
p_π (three regions)	GENIE multisim u545	0.831	23.5%	-0.45σ
	$\Delta \to N\pi$ angular	1.101	16.9%	$+0.41\sigma$

One bin deviates by more than one standard deviation of its uncertainty: the largest bias is 1.21σ, in the third FHC $\cos\theta_\mu$ bin under the multisim variation. In the well-populated bins the extraction follows the alternative truth: the five lower p_μ bins return 0.88–1.04 of it under both variations, with biases below 0.5σ, against truth-level changes of up to 6% in FHC and 3% in RHC.

The bins that do not follow it are sparse ones. The two near-threshold p_π regions return 0.49 and 0.43 of the alternative truth under the multisim variation, which lowers that truth by 18–20%; the open highest p_μ bin returns 1.28 under the angular variation; and the first and third FHC $\cos\theta_\mu$ bins ($\cos\theta_\mu < 0.4$ and 0.625–0.75) return 0.73 and 0.56 under the angular variation and 0.74 and 0.46 under the multisim one. These are the weakest bins of the measurement: the first two because the range-based pion momentum saturates, the others because they lie in the sparse tails. Their biases stay near or below 1σ partly because their uncertainties are large, 126–139%, 28% and 71–96%. At the present exposure the model bias is therefore inside the uncertainties, but it would become significant if the uncertainties were substantially reduced, and the test should be repeated for a larger exposure. In RHC no bin exceeds 0.90σ: the multisim variation returns 0.68–0.98 of the alternative truth in every p_μ bin (-0.1 to -0.90σ) and 0.82–0.86 on average in p_μ, $\cos\theta_\pi$ and p_π, while $\cos\theta_\mu$ comes back 1% low; under the angular variation the RHC ratios are 0.94–1.10. The same test for the proton-tagged extractions is reported in the companion note (Sec. 5.3 of the proton-tagged note).

Independent-generator closure. Reweighting reaches only changes of the parameters of the nominal model. A generator with a different resonance model, non-resonant treatment and final-state interactions, simulated through the detector, would test changes that reweighting cannot reach. No such test has been made. The one existing NuWro overlay, for Run-1 FHC, cannot serve: its true signal rate per POT is 1.76 times that of GENIE, against 1.6 expected from the standalone NuWro cross section and the flux, and its production could not be traced, so a closure test with it would test the sample rather than the extraction.

14 Results

14.1 Result set

The Primary results are the fifteen differential extractions, five observables in three configurations, and the three one-bin totals; the two-dimensional $\cos\theta_\pi \times \cos\theta_\mu$ in the combined configuration is a Secondary result (§14.7). Each carries the official covariance, detector variations and the muon-momentum-scale term included; the effects outside it are listed in §12.4. The detector term of the inclusive results is 12–19% (25% for FHC p_π). The proton-tagged extractions are reported in the companion note.

Table 19 gives, for the FHC results, the number of bins, the smallest migration diagonal, the range of the A_C row sums, the trace of A_C, which is the number of degrees of freedom that the data determine, the number of components measured with a signal-to-noise ratio of at least one, and the bin-averaged total uncertainty. A row sum well below one means that the corresponding bin is determined largely by the regularisation.

Table 19: Information content of the primary FHC results. The reported points are the Wiener-SVD estimator $A_C\, x_{\text{true}}$, not an unsmeared estimate of x_{true}; an A_C row sum well below one means that bin is set largely by the regularisation. The trace of A_C is the number of degrees of freedom the data determine, and the next column the number of components measured with signal-to-noise of at least one (Wiener filter factors ≥ 0.5); the uncertainty is the bin-averaged total including data statistics. Migration diagonals are the smallest over bins.

Observable	bins	min. diagonal	A_C row sums	trace A_C	factors ≥ 0.5	unc. [%]
p_μ	6	0.57	0.48–1.03	2.20	2	38.6
p_π (three regions)	3	0.52	0.42–0.83	1.58	2	52.1
$\cos\theta_\mu$	8	0.65	0.50–0.73	4.26	6	40.7
$\cos\theta_\pi$	5	0.76	0.66–0.76	3.15	4	38.6
$\theta_{\mu\pi}$	5	0.78	0.54–1.14	2.24	2	34.5

14.2 Differential cross sections

Tables 20–22 summarise the differential results of the three configurations, and Figs. 23–25 show them. All are extracted from pseudo-data, and all recover the A_C-smeared truth of the pseudo-data in every observable.

Table 20: FHC (Runs 1, 2, 4, 5; data POT 7.766×10^{20}): integrated cross section, the two largest systematic terms, the systematic total (all terms except the data statistics) and the closure against the A_C-smeared truth of the pseudo-data, from the covariance and result files. The flux term is discussed in §12.2.

Observable	σ_{int}	Flux	Detector	Syst. total	Closure χ^2/ndf
p_μ	0.874	28.6%	16.1%	37.0%	1.20/6 ($p = 0.98$)
p_π	0.843	36.5%	24.9%	49.4%	0.45/3 ($p = 0.93$)
$\cos\theta_\mu$	0.724	28.0%	18.8%	36.7%	3.12/8 ($p = 0.93$)
$\cos\theta_\pi$	0.655	27.1%	17.6%	35.8%	1.28/5 ($p = 0.94$)
$\theta_{\mu\pi}$	0.893	28.3%	13.5%	33.5%	0.52/5 ($p = 0.99$)

The RHC flux is 54.6% $\bar{\nu}_\mu$, against 35% in FHC (Table 6). The selected RHC events are nevertheless 76% ν_μ and 20% $\bar{\nu}_\mu$, since the ν_μ cross section is two to three times larger. The RHC overlays (Runs 1, 2, 3, 4a, 4b and 4c) were checked to be ν_μ overlays, with a truth composition of 100% $\nu_\mu + \bar{\nu}_\mu$ and the same number of events per POT in every file (2.62×10^{-16} with the PPFX weights); together they contain 4.05×10^6 events and 1.546×10^{22} simulated POT, the largest period being Run 3 (5.52×10^{21} POT, 1.44×10^6 events). The RHC detector uncertainty uses the native Run-4 RHC detector-variation samples (light yield down, Rayleigh scattering, recombination, space charge and four wire-response variations), with 3.80×10^{-16} events per POT in every variation. These samples, like the FHC ones, were simulated with the earlier NuMI flux: weighted event by event to the current flux their rate is 0.94 times that of the nominal overlays in both modes, and the ratio no longer rises with energy. The detector term therefore uses each variation relative to its own CV sample (§12). The dirt background uses the RHC dirt overlays of Runs 3b, 4a and 4b, with the Run-3b sample standing in for Runs 1 and 2, each scaled to the exposure of its run. The generator predictions are averaged over the RHC flux composition.

The combined measurement uses the summed response of the fourteen overlay files, 5.6×10^6 events, with each horn mode normalised to its own exposure (1.885×10^{21} POT in total). The prediction thus weights the two modes by their recorded exposure, 0.70:1, rather than by their simulated exposure, 0.62:1, and the pseudo-data are thrown at the same rates. The combined generator predictions are the fluence-weighted average of the two modes, 0.425 FHC and 0.575 RHC, each mode weighted by $\Phi_{\text{tot}} \times \text{POT}$.

The flux uncertainty of the combination needs no separately constructed correlation between the horn modes. All fourteen overlay files carry the same 600 PPFX multisim universes, and the universes are accumulated index by index across both modes, so universe u is the same hadron-production variation in FHC and RHC. Summing per universe propagates each variation through both configurations and yields the physical correlation between the modes. The detector variations are carried per mode, with the Run-4 FHC and Run-4 RHC samples of each variation accumulated under one label, so that the combined detector term is weighted by exposure like the central value.

The combined and RHC closures pass in every observable (Tables 21 and 22): the RHC truth is recovered with $\chi^2 \leq 1.62$ on three to eight degrees of freedom and $p \geq 0.99$. Excluding p_π, the unfolded flux term is 24.1–26.6% combined, 27.1–28.6% in FHC and 25.9–27.9% in RHC (31.8%, 36.5% and 29.2% for p_π), as expected for a common flux model propagated through similarly conditioned responses. The reconstructed-level flux term is about 15% in RHC and 17% in FHC, the $\bar{\nu}_\mu$-dominated flux having a smaller hadron-production uncertainty, but the unfolded terms are similar because the background subtraction multiplies both by $1 + B/S$ (§12.2).

Table 21: RHC (Runs 1, 2, 3, 4; data POT 1.1082×10^{21}): integrated cross section, the two largest systematic terms, the systematic total (all terms except the data statistics) and the closure against the A_C-smeared truth of the pseudo-data, from the covariance and result files. The flux term is discussed in §12.2.

Observable	σ_{int}	Flux	Detector	Syst. total	Closure χ^2/ndf
p_μ	0.688	27.9%	19.3%	38.8%	0.30/6 ($p = 1.00$)
p_π	0.743	29.2%	14.4%	37.5%	0.01/3 ($p = 1.00$)
$\cos\theta_\mu$	0.710	25.9%	17.4%	34.0%	1.62/8 ($p = 0.99$)
$\cos\theta_\pi$	0.615	25.9%	16.9%	33.7%	0.52/5 ($p = 0.99$)
$\theta_{\mu\pi}$	0.806	26.2%	12.5%	31.8%	0.09/5 ($p = 1.00$)

Table 22: Combined (data POT 1.8848×10^{21}): integrated cross section, the two largest systematic terms, the systematic total (all terms except the data statistics) and the closure against the A_C-smeared truth of the pseudo-data, from the covariance and result files. The flux term is discussed in §12.2.

Observable	σ_{int}	Flux	Detector	Syst. total	Closure χ^2/ndf
p_μ	0.804	26.3%	17.6%	35.6%	0.22/6 ($p = 1.00$)
p_π	0.778	31.8%	17.4%	41.5%	0.13/3 ($p = 0.99$)
$\cos\theta_\mu$	0.750	24.9%	16.1%	32.1%	1.91/8 ($p = 0.98$)
$\cos\theta_\pi$	0.665	24.1%	13.9%	31.1%	0.95/5 ($p = 0.97$)
$\theta_{\mu\pi}$	0.844	26.6%	12.1%	31.7%	0.18/5 ($p = 1.00$)

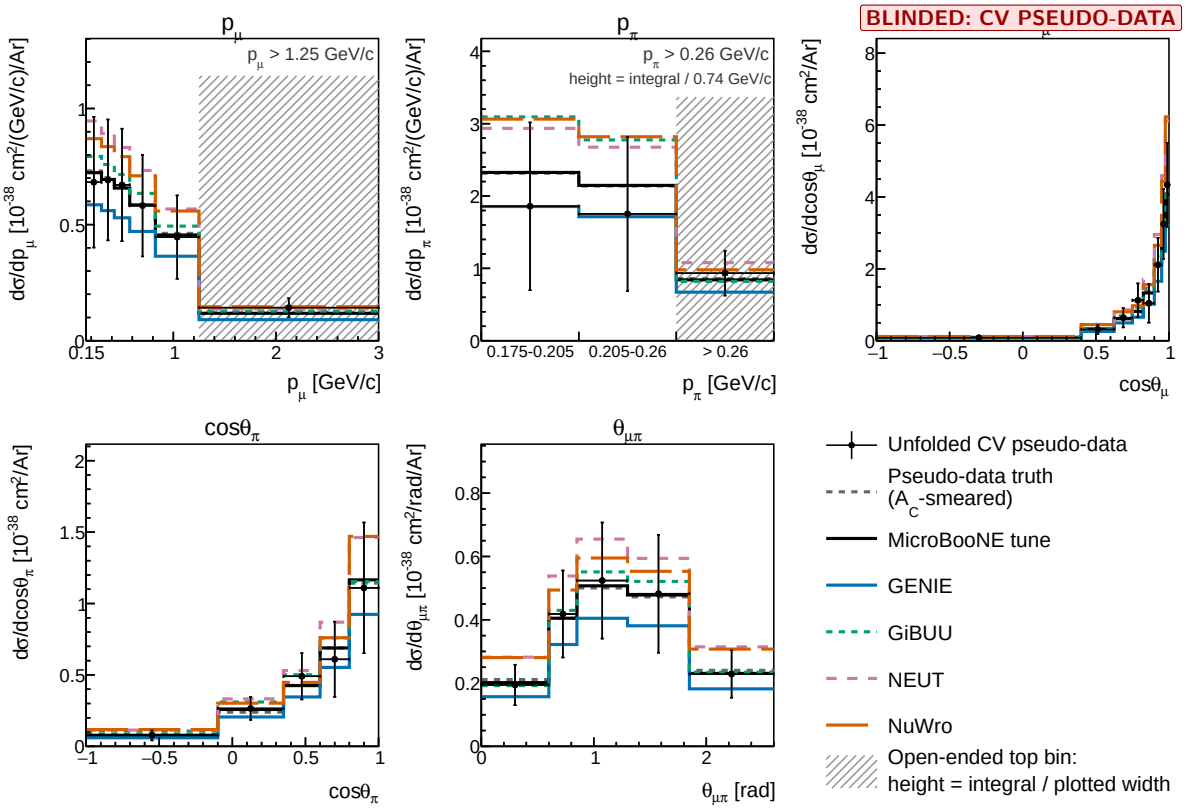


Figure 23: Differential cross sections $d\sigma/dx$ of the five distributions, FHC full exposure, extracted from pseudo-data. The points are the Wiener-SVD estimator $A_C\, x_{\text{true}}$; shown are the unfolded pseudo-data (black points), the A_C-smeared truth of the pseudo-data (grey dashed), the MicroBooNE tune (black solid) and the four generator predictions (GENIE, GiBUU, NEUT, NuWro), all smeared with A_C. The unfolded points recover the injected truth in every observable, the closure summarised in Table 20.

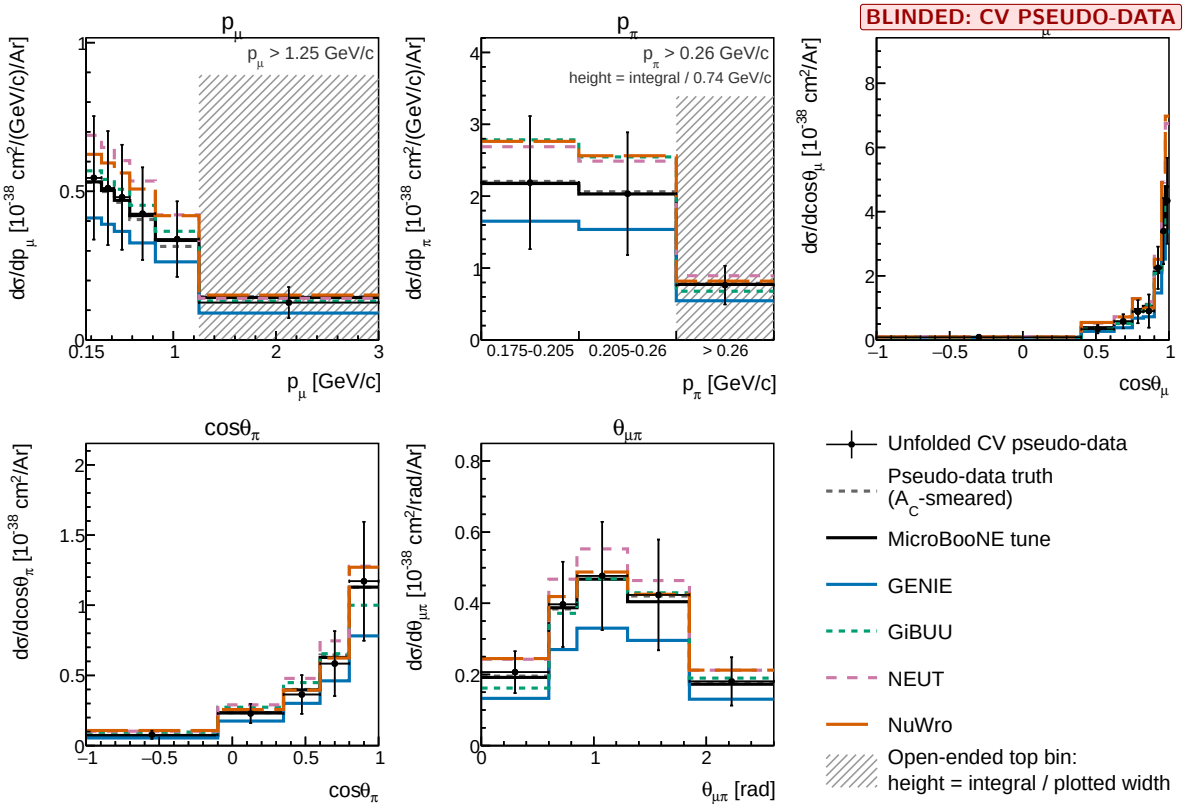


Figure 24: RHC differential cross sections from pseudo-data, with the generator predictions averaged over the RHC flux. Same layout as Fig. 23.

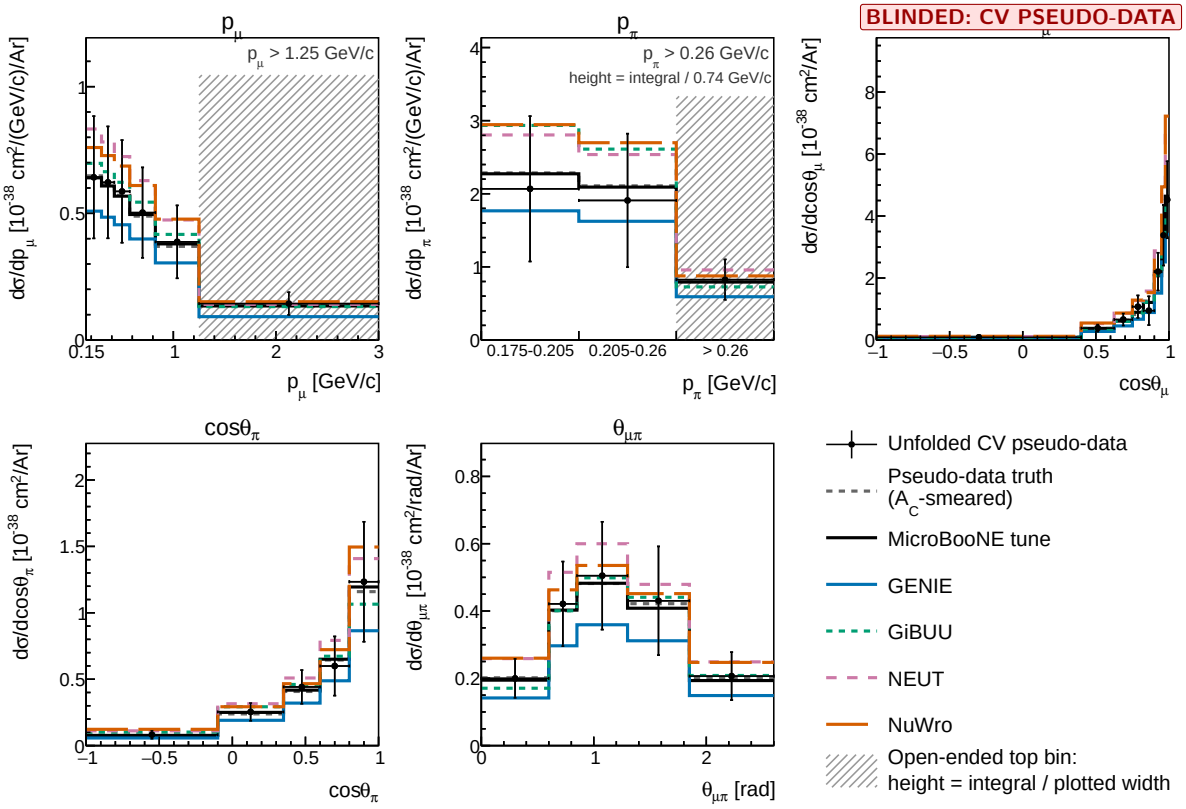


Figure 25: Combined FHC+RHC differential cross sections from pseudo-data normalised per horn mode, with the fluence-weighted combination of the generator predictions. Same layout as Fig. 23.

Table 23: Integrated cross section σ_{int} of each unfolded differential result over the measured phase space (10^{-38} cm^2/Ar), from pseudo-data, with the $\nu_\mu + \bar{\nu}_\mu$ flux of each configuration integrated above 60 MeV. The value differs between observables because A_C does not preserve the normalisation; the D'Agostini cross-check (Sec. 12 of the technical supplement) gives a total 15–59% higher, independent of the observable to 1% and equal to the one-bin total, and the A_C row sums of §14.4 show the effect directly. The total cross section is the one-bin extraction of Table 27.

Observable	FHC	RHC	Combined
p_μ	0.874	0.688	0.804
p_π	0.843	0.743	0.778
$\cos\theta_\mu$	0.724	0.710	0.750
$\cos\theta_\pi$	0.655	0.615	0.665
$\theta_{\mu\pi}$	0.893	0.806	0.844

Uncertainties shown. The error bars of the unfolded points are the square roots of the diagonal of the total covariance propagated through the unfolding: flux (PPFX multisims), cross section (GENIE multisims and unisims), detector variations [15], hadron re-interaction [25], POT and target number, the muon momentum scale, and the data, simulation and beam-off statistical terms. For FHC p_μ the bin-averaged total is 38.6% (37.0% without the data statistics, Table 17), with flux 28.6%, detector 16.1%, data statistics 10.9%, cross section 11.0% and muon momentum scale 10.1%. The one-bin total (§14.6) has its own breakdown from the same universes, systematic $\oplus$ statistical $= 35.6\% \oplus 4.4\%$ in FHC; without regularisation its flux and detector terms are not redistributed between bins.

14.3 Pion momentum in three regions

The upper pion-momentum bin is open-ended, so the p_π result is reported as three bin-integrated cross sections (Table 24): two intervals near threshold, 0.175 to 0.205 and 0.205 to 0.26 GeV/c, and everything above. These are the quantities to compare with a prediction. The differential form, with a conventional width for the upper bin (Appendix A), is used only to draw the result together with the other observables.

Table 24: The p_π result as bin integrals (10^{-38} cm^2/Ar), from pseudo-data. The upper region is open-ended and has no width. The uncertainties are the total of the covariance. All three bins are A_C-smeared, so a prediction is compared after multiplication by A_C.

Region	FHC	RHC	Combined
$\sigma(0.175 < p_\pi < 0.205\,\mathrm{GeV}/c)$	0.056 ± 0.035	0.066 ± 0.028	0.062 ± 0.030
$\sigma(0.205 < p_\pi < 0.26\,\mathrm{GeV}/c)$	0.096 ± 0.059	0.112 ± 0.047	0.105 ± 0.050
$\sigma(p_\pi > 0.26\,\mathrm{GeV}/c)$	0.691 ± 0.229	0.565 ± 0.199	0.611 ± 0.205

14.4 Comparing a prediction with the results

The Wiener-SVD result estimates $A_C x_{\mathrm{true}}$, and A_C does not preserve the normalisation, so the sum of a differential result over its bins is not the total cross section. The row sums of A_C (Table 25) describe the effect: a row summing to one passes the normalisation of a flat bin-integrated truth unchanged, and a row summing to 0.2 passes one fifth of it. For a non-flat prediction, bin i receives $\sum_j (A_C)_{ij}\, t_j$, which depends on the individual elements and on the shape of t. For FHC p_μ the ratio of the smeared integral to the true total is 0.80, 0.80, 0.78 and 0.81 for GENIE, GiBUU, NEUT and NuWro. The true totals of the same generators span 0.82 to 1.32×10^{-38} cm^2/Ar (Table 19 of the technical supplement), a factor 1.6, so a common factor near 0.80 does not produce the differences between the generators; they are model differences.

Table 25: Range of the row sums of A_C for each observable and configuration, from the result files.

Observable	FHC	RHC	Combined
p_μ	0.48–1.03	0.39–0.75	0.49–0.81
p_π	0.42–0.83	0.40–0.94	0.38–0.88
$\cos\theta_\mu$	0.50–0.73	0.44–0.79	0.51–0.71
$\cos\theta_\pi$	0.66–0.76	0.71–0.81	0.56–0.77
$\theta_{\mu\pi}$	0.54–1.14	0.59–1.13	0.65–1.22

The suppression is large, down to a row sum of 0.38 for p_π combined and 0.39 for p_μ in RHC, and differs strongly between observables. This is why the integrated cross section of a differential result varies between observables while the physical total cannot; the D'Agostini cross-check, whose integral does not depend on the observable, agrees with this interpretation (Sec. 12 of the technical supplement).

14.5 Model comparisons

Table 26 gives the χ^2 of the unfolded pseudo-data against each prediction, for every observable and configuration. All predictions are smeared with A_C, so the comparison is made in the space of the measurement, with the full covariance. The truth column is the closure test.

Every prediction is smeared with A_C exactly once, by the extraction, and the figures use the smeared curves without smearing again. The prescription above, applied to the true predictions, reproduces the smeared curves to a relative 4×10^{-8} (Sec. 22 of the technical supplement).

A second application would over-smooth the sharply peaked generators and create a spurious separation in the muon-angle observables. With a single application NEUT and NuWro give 4.36/8 and 5.56/8 in FHC $\cos\theta_\mu$, consistent with the other observables.

Table 26: χ^2/ndf of the unfolded pseudo-data against each A_C-smeared prediction, for the inclusive selection. "truth" is the pseudo-data truth (the closure test).

Beam	Observable	truth	uB tune	GENIE	GiBUU	NEUT	NuWro
FHC	p_μ	1.20/6	2.14/6	3.47/6	2.09/6	3.32/6	1.92/6
	p_π	0.45/3	0.56/3	1.03/3	2.58/3	0.98/3	1.34/3
	$\cos\theta_\mu$	3.12/8	2.44/8	3.36/8	3.22/8	4.36/8	5.56/8
	$\cos\theta_\pi$	1.28/5	0.99/5	1.58/5	2.34/5	2.33/5	3.16/5
	$\theta_{\mu\pi}$	0.52/5	0.47/5	0.94/5	0.75/5	3.24/5	3.91/5
RHC	p_μ	0.31/6	0.47/6	0.93/6	0.44/6	0.68/6	1.09/6
	p_π	0.01/3	0.01/3	0.68/3	1.17/3	0.35/3	0.43/3
	$\cos\theta_\mu$	1.62/8	1.95/8	2.32/8	1.09/8	8.28/8	13.49/8
	$\cos\theta_\pi$	0.52/5	0.49/5	1.56/5	3.19/5	1.89/5	1.69/5
	$\theta_{\mu\pi}$	0.09/5	0.43/5	2.49/5	3.64/5	0.81/5	1.37/5
comb	p_μ	0.22/6	0.34/6	1.57/6	0.37/6	1.24/6	0.59/6
	p_π	0.13/3	0.16/3	0.84/3	2.38/3	0.61/3	1.07/3
	$\cos\theta_\mu$	1.91/8	1.71/8	2.83/8	1.76/8	7.43/8	13.37/8
	$\cos\theta_\pi$	0.95/5	0.65/5	1.79/5	3.46/5	2.31/5	3.02/5
	$\theta_{\mu\pi}$	0.18/5	0.52/5	1.85/5	2.51/5	2.84/5	3.79/5

14.6 Total cross section

The total flux-averaged cross section in the measured phase space is extracted with the same machinery as a one-bin distribution: a single true bin that contains every signal event, defined on $\cos\theta_\mu$ over its full range $[-1, 1]$ so that no overflow exists, and a single reconstructed bin over the same range. With one bin the response $S = \varepsilon M$ reduces to the efficiency, nothing is regularised, the Wiener filter is switched off, and the unfolding is the inversion

$$\sigma = \frac{N_{\text{sel}} - N_{\text{bkg}}}{\epsilon\, \Phi\, N_{\text{Ar}}}, \tag{16}$$

with $A_C \equiv 1$ exactly. Here N_{sel} is the selected count, N_{bkg} the predicted background (neutrino background, beam-off and dirt), $\epsilon = N^{\text{sig}}_{\text{sel}}/N^{\text{sig}}_{\text{gen}}$ the selection efficiency in the measured phase space (18.3% in FHC, 18.4% in RHC and combined), Φ the integrated flux ($\Phi = \text{POT} \times \Phi_{\text{POT}}$, Table 6) and $N_{\text{Ar}} = 8.71 \times 10^{29}$ the number of argon nuclei in the fiducial volume, the same normalisation $N_{\text{Ar}}\,\Phi/10^{-38}$ as for the differential results. Every term of Eq. (16) is evaluated universe by universe, so the covariance contains the flux, cross-section, re-interaction and detector variations, the POT and target terms and the simulation, beam-off and data statistical terms, propagated as for the differential results. The flux term is 28–30%, the flux normalisation times $(1 + B/S)$ with B the neutrino background (§12.2); the detector term is 14% and the data-statistical term 3–4%. The differential results carry the same terms bin by bin, with the regularisation redistributing them between bins. Without additional smearing the one-bin result is the physical total. It lies above the A_C-suppressed σ_{int} of Table 23, most visibly in RHC, and agrees with the D'Agostini cross-check (Sec. 12 of the technical supplement).

A closed-form evaluation of Eq. (16) on the central-value prediction checks the normalisation chain: it reproduces the one-bin extraction of the central-value prediction to four digits (1.0315, 0.9285 and 0.9723 for FHC, RHC and combined) and provides the N_{sel}, N_{bkg} and ϵ columns of Table 27. The counts carry the full central-value weight (tune × PPFX × normalisation, §5);

the PPFX weight averages 0.944 on selected signal, so a count without it would lie 3–6% above the prediction used in the unfolding.

Table 27 gives the total cross sections. The σ column is the one-bin extraction from pseudo-data and fluctuates around the tune by the Poisson throw; the realised truth of the throw is listed, and the ratio to it is 0.990–0.991 with $\chi^2/1 \ll 1$ in the three configurations. The uncertainty is that of the one-bin extraction itself. The ratio RHC/FHC of the tune is 0.900; the extracted ratio from the pseudo-data is 0.901.

Table 27: Total cross sections (10^{-38} cm^2/Ar) in the measured phase space from the one-bin extraction (§14.6) on central-value pseudo-data, averaged over the $\nu_\mu + \bar{\nu}_\mu$ flux above 60 MeV, which includes the flux below the pion-production threshold. N_{sel} and N_{bkg} (neutrino background + beam-off + dirt) are the selected count and background of the central-value prediction and ε the selection efficiency; "stat" is the data-statistical term, "syst" all other terms and "total" their quadrature sum; "truth" is the realised truth of the throw and $\chi^2/1$ the closure against it. The counts agree with Table 11.

Config	N_{sel}	N_{bkg}	ε [%]	σ	stat [%]	syst [%]	total [%]	truth	σ/truth	$\chi^2/1$
FHC	1 522	651	18.3	1.042 ± 0.374	4.4	35.6	35.9	1.044	0.998	0.000
RHC	1 881	817	18.4	0.925 ± 0.316	4.1	33.9	34.1	0.941	0.982	0.003
Combined	3 404	1 467	18.4	0.975 ± 0.337	3.0	34.5	34.6	0.985	0.990	0.001

The flux-averaged generator totals in the inclusive phase space are 0.82 (GENIE), 1.13 (GiBUU), 1.32 (NEUT) and 1.25×10^{-38} cm^2/Ar (NuWro) for FHC (Table 19 of the technical supplement for all three configurations). The one-bin total lies 0.6–0.8σ above GENIE and within 0.7σ of the other three; the spread of the generators, a factor 1.6, is larger than any of these differences.

14.7 Two-dimensional cross section

The double-differential cross section $d^2\sigma/d\cos\theta_\pi\, d\cos\theta_\mu$ is a result proposed for the combined configuration only. It is binned in two regions of $\cos\theta_\pi$ (edges -1, 0.35, 1) inside three of $\cos\theta_\mu$ (-1, 0.65, 0.85, 1); the six analysis bins are ordered outer slice by outer slice and extracted with the same machinery as the one-dimensional results, a two-dimensional binning being a flat list of analysis bins. The pair is one of eight screened (Sec. 10 of the technical supplement). Combining the beam modes raises the conditioning $s_{\min}/s_1$ of its A_C from 0.301 (FHC) and 0.127 (RHC) to 0.319, through the larger sample and the shared flux; the FHC and RHC extractions cannot be inverted and are Not reported. The combined result has a bin-averaged data-statistical term of 15.4% and per-bin total uncertainties of 29–58%; it closes against the A_C-smeared truth of the pseudo-data with $\chi^2 = 0.76/6$, with per-bin ratios of 0.93–1.33 (Figs. 26 and 27). Its covariance, A_C and curves are released as `incl_COMB_costhpi_costhmu2d`.

The result was validated as the one-dimensional ones were. Over 100 pseudo-data sets (§13.2) the statistical pull width is 1.03 and the 68% and 95% coverages are 68% and 93%, so the statistical uncertainties are correctly sized. The ensemble mean lies below the reference in the slice $\cos\theta_\mu < 0.65$: by 12% in the cell with $\cos\theta_\pi < 0.35$, the sparsest of the six, which is 0.57 of its statistical and 0.18 of its total uncertainty, and by 2% in the other cell. The first $\cos\theta_\mu$ bin of the one-dimensional ensembles shows the same effect at a smaller size, -0.25 (FHC) and -0.47 (RHC) statistical standard deviations. Through these two cells the ensemble-mean integral lies 0.8% below the reference, the size of the common offset of the one-dimensional ensembles; it amounts to 2.4 standard errors of the mean because the integral of this result varies by only 3.3% from one pseudo-data set to the next. No correction is applied: added in quadrature, the offset of the sparsest cell would raise its total uncertainty from 52% to 53%. Under the two alternative models of §13.3 the result follows the alternative truth to within 0.62σ in every cell. The GENIE

variation changes the true cells by -23% to $+16\%$, and the result follows it to within 0.38σ; under the angular variation the cell with $\cos\theta_\pi < 0.35$ and $0.65 < \cos\theta_\mu < 0.85$ returns 0.79 of the alternative truth (-0.62σ).

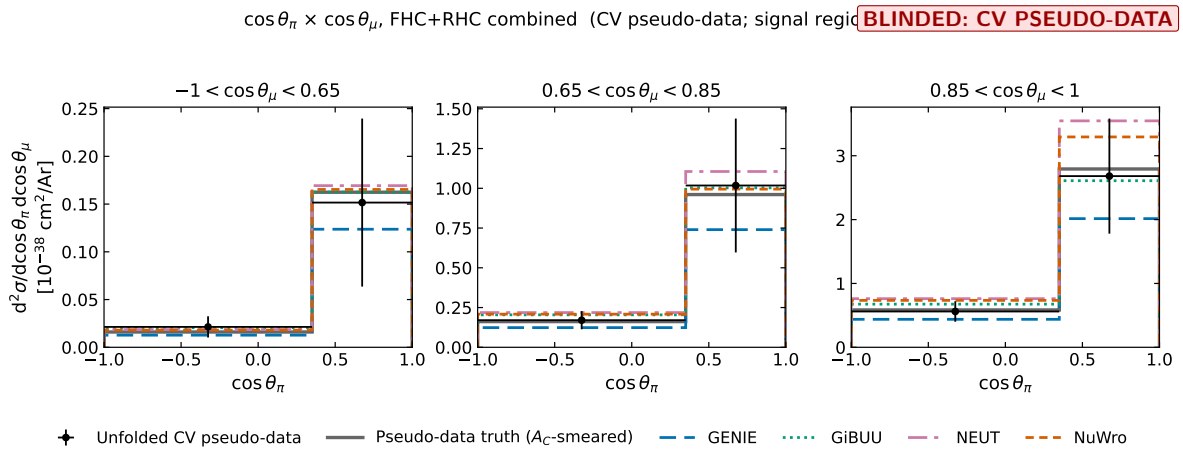


Figure 26: $d^2\sigma/d\cos\theta_\pi\, d\cos\theta_\mu$ in the combined configuration, as slices in $\cos\theta_\mu$. Points are the unfolded pseudo-data with the total uncertainty, the grey histogram is the A_C-smeared truth of the pseudo-data and the coloured histograms are the generator predictions, smeared with A_C. Each slice has its own vertical scale.

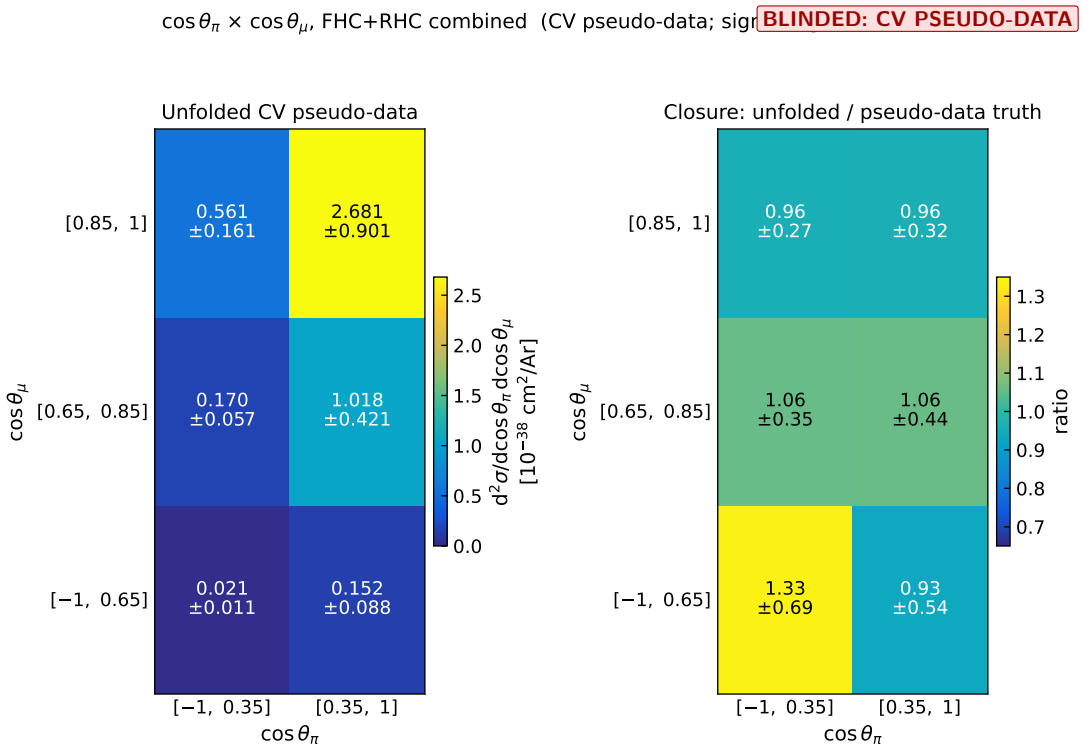


Figure 27: The same result as a map: $d^2\sigma/d\cos\theta_\pi\, d\cos\theta_\mu$ per cell with its total uncertainty (left) and the ratio to the A_C-smeared truth (right). Cells are drawn with equal size and labelled by their ranges.

14.8 Proton-tagged subsample

The proton-tagged subsample, the inclusive selection with a reconstructed proton candidate above 0.3 GeV/c, is measured separately [45]: a one-bin total, the proton momentum, the calorimetric hadronic mass, four transverse-kinematic-imbalance observables, two proton angles and two-dimensional pairs. It has its own signal definition, proton-assignment requirements and validation.

15 Limitations

Blind signal region. The signal region has not been examined in beam-on data, and every result is extracted from pseudo-data. The results therefore demonstrate the extraction and its uncertainties, not the cross section. The background model enters strongly at 57–59% purity, with a proton reconstructed as the pion as the largest neutrino background, and closure tests cannot probe it because the same simulation provides the pseudo-data and the background estimate. The four control regions are defined to test it at reconstruction level with beam-on data (§10).

Pion momentum. Charged pions interact before they stop, so the reconstructed momentum saturates near 0.34 GeV/c and carries little information above about 0.4 GeV/c. Golden-pion tagging, multiple Coulomb scattering, calorimetry, multivariate regression, recovery of the daughter energy and the proton-tagged selection were all studied, and a truth-level golden selection performs no better than the trained classifier, which places the limit in the physics of pion interactions rather than in the estimator. For this reason p_π is reported in three regions.

Model dependence. The closure tests use pseudo-data from the model of the response. The test with reweighted interaction models bounds the model bias at 1.21σ (§13.3), but a test with an independent generator and its own detector simulation has not been made. Reweighting probes only model changes expressible through the event weights; it does not change final-state multiplicities or kinematic correlations outside the support of the nominal simulation. The model dependence of the efficiency matters most for the one-bin total, a single efficiency averaged over the phase space, and for the open-ended $p_\pi > 0.26$ GeV/c region, inside which the efficiency varies strongly with the true momentum.

Muon momentum scale. The MCS momentum-scale term (§12.3) is part of the official covariance. Its size, $\pm 5\%$, is assumed and has not been compared with the difference between the MCS and range momenta of contained muons in data.

Proton-tagged results. The limitations specific to the proton-tagged subsample (the proton mis-assignment, which no control region yet constrains, and the transverse-kinematic prescription) are discussed in the companion note [45].

16 Summary

We have developed a measurement of $\nu_\mu + \bar{\nu}_\mu$ charged-current single charged-pion production on argon with the full MicroBooNE NuMI exposure: forward horn current (7.766×10^{20} POT), reverse horn current (1.1082×10^{21} POT) and their combination (1.8848×10^{21} POT). The signal region is blind, and all results are extracted from pseudo-data thrown from the central-value simulation; no statement about a generator is yet a statement about nature.

Phase space and selection. The signal is a ν_μ or $\bar{\nu}_\mu$ charged-current interaction with its vertex in the fiducial volume and exactly one muon and one charged pion, no neutral pion or kaon and any number of nucleons, with $p_\mu > 0.15\,\mathrm{GeV}/c$, $p_\pi > 0.175\,\mathrm{GeV}/c$ and $\theta_{\mu\pi} < 2.6$ rad. The selection reaches 18.3–18.4% efficiency at 57% purity. Angles are measured about the neutrino direction of each event, 28.6° from the detector axis.

Results. In each configuration the total flux-averaged cross section and five differential distributions are measured: $d\sigma/dp_\mu$, $d\sigma/d\cos\theta_\mu$, $d\sigma/d\cos\theta_\pi$, $d\sigma/d\theta_{\mu\pi}$, and p_π in three regions; these and the total are the Primary results. The combined-configuration $d^2\sigma/d\cos\theta_\pi\, d\cos\theta_\mu$ is a Secondary result. The binnings meet a 0.50 migration criterion (§11.3). The cross sections are averaged over the $\nu_\mu + \bar{\nu}_\mu$ flux integrated above 60 MeV. The total cross section carries a 34–36% uncertainty, with flux 28–30% and detector 14%; the flux term is the 17% flux normalisation multiplied by $1 + B/S$ through the background subtraction. The differential results carry 31–42% bin-averaged uncertainty without the data statistics (49% for FHC p_π), including the muon-momentum-scale term. Each differential result is provided with its additional-smearing matrix A_C, which must be applied to a prediction before comparison, and with its covariance by source.

Background. The background, 42–43% of the selected sample, is taken from the simulation and from beam-off data. Four control regions, disjoint from the signal region, are defined to test the prediction; they have been studied with pseudo-data, and their comparison with beam-on data has not been made.

Validation. Closure recovers the input in every extraction, with a mean ratio of 1.001 over the fifteen inclusive extractions. Ensembles of 100 pseudo-experiments in seven extractions give pull widths of 0.92–1.04 and correct coverage in all three configurations. Pseudo-data from reweighted interaction models bias no bin by more than 1.21σ (§13.3). The identity $UR = A_C$ holds to 5.7×10^{-5} (App. 13 of the technical supplement), and the A_C row sums show that the difference between the integrals of the differential results is the additional smearing. The result set is rebuilt from the processed ntuples with empty caches in a single pass; an independent rebuild into a separate tree reproduced every bin of the fifty-one extractions to 10^{-9} (Sec. 22 of the technical supplement), and the generator curves are reproduced from the provided A_C to a relative 4×10^{-8}, the precision of the text files.

Limitations. The saturation of the range-based pion momentum near $0.34\,\mathrm{GeV}/c$ limits p_π to three regions and is the main physics limitation of the selection. The control regions have not been compared with beam-on data, and no closure test with an independent generator has been made (§15).

Appendix

A Observable binning

The binnings are the same in the three configurations and meet the 0.50 migration criterion in FHC and RHC. They were selected in the binning review of Sec. 7 of the technical supplement: among the binnings that meet the criterion and hold at least 70 selected FHC signal events in every true bin, the one with the fewest bins that reaches 90% of the largest trace of A_C, the number of degrees of freedom that the data determine. $\cos\theta_\pi$ and $\theta_{\mu\pi}$ keep five bins, which pass both criteria; the seven and six bins the rule selects would add only 0.45 and 0.1 degrees of freedom. The generator predictions are computed at these edges from the event-level files.

Table 28: Bin edges of every observable, from the result files. The binnings meet the 0.50 migration criterion (Sec. 7 of the technical supplement) and are the same in the three configurations.

Observable	Bins	Edges
p_μ (GeV/c)	6	0.15, 0.3, 0.425, 0.575, 0.825, 1.25, 3*
p_π (GeV/c)	3	0.175, 0.205, 0.26, 1*
$\cos\theta_\mu$	8	-1, 0.4, 0.625, 0.75, 0.825, 0.9, 0.95, 0.975, 1
$\cos\theta_\pi$	5	-1, -0.1, 0.35, 0.6, 0.8, 1
$\theta_{\mu\pi}$ (rad)	5	0, 0.6, 0.85, 1.3, 1.85, 2.6

* Numerical plotting boundary only: the upper truth bin is open-ended, and a density drawn over the conventional width represents the bin integral.

The upper p_π region is defined at truth level as $p_\pi > 0.26$ GeV/c with no upper edge. An open-ended bin has no width and hence no differential density: the 1 GeV/c in Table 28 is a convention that assigns a width of 0.74 GeV/c so that the region can be drawn on the same axis as the other two, not a kinematic limit of the sample. A height read from that figure reflects the convention. The three region integrals are unambiguous, and the result is quoted and compared as those integrals (Table 24). The other open-ended bins of the table follow the same convention.

B Alternative binnings

The binning review (Sec. 7 of the technical supplement) compares the adopted binnings with those the same rule selects under the stricter 0.68 criterion: four p_μ bins, two p_π regions and seven $\cos\theta_\mu$ bins. Table 7 of the technical supplement compares the two criteria observable by observable. The binnings of an earlier optimisation (Sec. 23 of the technical supplement) are not used: its seven-bin p_μ fails both criteria, with a worst diagonal of 0.33, and its five-bin $\cos\theta_\mu$ fails the 0.68 criterion narrowly (0.67). Table 29 lists the other inclusive binnings examined, with the conditioning $s_{\min}/s_1$ of A_C on which they were first judged; the review replaced that measure by the trace of A_C, since $s_{\min}/s_1$ is set by the weakest component alone and most binnings, the adopted ones included, do not meet it.

Table 29: Alternative inclusive binnings and their status. $s_{\min}/s_1$ is the conditioning of A_C in FHC/RHC.

Observable	Alternative	$s_{\min}/s_1$	Status
$\cos\theta_\mu$	6 bins	0.11/0.09	superseded by the eight bins selected in the review
	7 bins	0.02/0.05	superseded by the eight bins selected in the review
θ_μ	7, 8 bins	0.00/0.00	not adopted: $\cos\theta_\mu$ used (Sec. 24 of the technical supplement)
$\cos\theta_\pi$	4 bins	0.27/0.24	superseded by the five bins with a forward split

C Selection cut parameters

Table 30: All selection threshold constants.

Parameter	Value	Ntuple variable
Topological score	> 0.67	Pandora score
Muon track score	> 0.80	Pandora track score
Muon vertex distance	≤ 4.0 cm	
Muon track length	> 10.0 cm	
Muon LLR PID	> 0.20	LLR PID
Muon MIP BDT	≥ -0.10	TMVA BDT
Pion track length	> 20.0 cm	
Pion vertex distance	< 4.0 cm	
Pion LLR PID	> 0.10	LLR PID
Pion MIP BDT	> -0.10	TMVA BDT
Pion pion-BDT	> -0.10	TMVA BDT
Max. opening angle	< 2.6 rad	
Shower Y-plane hits	< 50 (if 1 shower)	
Shower vertex distance	> 10 cm (if 1 shower)	
Non-proton multiplicity	< 4	LLR PID > 0.1
Primary track multiplicity	< 5	
Cosmic BDT	> 0.0 (disabled)	TMVA BDT
Software trigger	$= 1$	

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